

TOPIC 12: DIVERSITY OF ORGANISMS

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (ATTITUDES, SKILLS AND KNOWLEDGE)	SUGGESTED LEARNING ACTIVITIES AND NOTES	SUGGESTED RESOURCES
8.15.1 Classification	- identify organisms using diagnostic features of the five Kingdoms - use diagnostic features to divide kingdoms into phyla - state the taxonomic hierarchy - observe the rules of binomial nomenclature	- Diagnostic features of the five Kingdoms - Diagnostic features of phyla - Kingdom - Phyla - Class - Order - Family - Genus - Species - Binomial nomenclature - Genus and species names	- Observing organisms. - Classifying organisms into the five Kingdoms. - Collecting and classifying organisms. - Outlining the taxonomic hierarchy. - Discussing the rules of binomial nomenclature.	- ICT tools - Braille software/ Jaws - Samples of organisms - Dichotomous key
8.15.2 Importance of Biodiversity	describe the socio-economic importance of the five Kingdoms	- socio-economic importance of I. Kingdom Prokaryotae - fermentation - bio-technology - food spoilage - decomposition II. Kingdom Protista - Plasmodium sp - - malaria <i>Schistosoma</i> sp - Schistosomiasis - Trypanosoma sp-Trypanosomiasis III. Kingdom Fungi - Fermentation - Penicillin production - Decomposition - Food spoilage - Food IV. Kingdom Plantae - Producers - Carbon sink - Timber - Medicinal use - Tourism V. Kingdom Animalia - Tourism - Food - Hunting - Leather - Fishing	Discussing the socio-economic importance of the five kingdoms.	- ICT tools - Brail software/ Jaws

CLASSIFICATION

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Classification is the placing of living organisms in a series of increasingly specialized groups. This is based on their similarities and difference in structure and origin that indicate a common relationship.

General Principles of classification

- Systems of classification are **hierchial** i.e. each successive group contains more and more different kinds of organisms
- Taxonomy is the study of grouping of organisms according to their relationship.
- A group of similar organisms is called a taxon, and the science of classification is called taxonomy.
- **Phylogeny** is the study of evolutionary traits
- Natural classification of organisms is based on evolutionary relationship
- In taxonomy groups are based on similar physical or molecular properties, and groups are contained within larger composite groups with no overlap.
- There are **seven taxons** that organisms can classified into:
 - Kingdom
 - Phylum (Plural: Phyla) or division in plants
 - Class
 - Order
 - Family
 - Genus
 - Species
- The smallest group of similar organisms is the species; closely related species are grouped into genera (singular genus), genera into families, families into orders, orders into classes, classes into phyla (singular phylum), and phyla into kingdoms.
- The **kingdom is the first and largest rank** and **species is the last and smallest**.
- So you need to remember **KPCOFGS**. **The mnemonic is King Philip Called Out For Genuine Scientists.**

DEFINITION OF TERMS

- **Kingdom** is the largest grouping of organisms' e.g. Animalia
- **Phylum** consists of organisms with many similarities e.g. bryophyte, cnidarians etc.
- **Class** consists of organisms which are grouped into several orders with few similarities
- **Order** is a group of apparently related families
- **Family** is a group of apparently related genera
- **Genus** is a group of similar and closely related species
- **A Species** is a group of organisms that can freely and naturally interbreed to give rise to a fertile (Viable) offspring.
An example is goats interbreed to produce goats because they belong to same species.
- In animals except human beings, the term breed is used while in plants the term variety is preferred.

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- As we go through the taxon hierarchy from kingdom to species, the groups get smaller and the animals are more closely related.
- This shows how the seven taxons are used to classify humans.

Kingdom	Animalia
Phylum	Chordata
Class	Mammalia
Order	Primates
Family	Hominidae
Genus	<i>Homo</i>
Species	<i>sapiens</i>

This shows the complete classification of some other species:

	Earthworm	Mushroom	Garlic
Kingdom	Animalia	Fungi	Plantae
Phylum	Annelida	Mycota	Angiospermophyta
Class	Oligochaeta	Basidiomycota	Monocotyledonea
Order	Terricolae	Agaricales	Liliales
Family	Lumbricidae	Agaricaceae	Liliaceae
Genus	<i>Lumbricus</i>	<i>Agaricus</i>	<i>Allium</i>
Species	<i>terrestris</i>	<i>campestris</i>	<i>sativum</i>

Binomial Nomenclature

This is a system of giving two **Latin** names to organisms. It was pioneered by a scientist called **Carolus Linnaeus** and the names given are **Generic** (genus) and **Specific** (Species).

Procedures followed

1. The **generic name** is written first followed by the **specific name**
2. The **generic name** starts with a capital letter and the rest are small letters. The **specific name** is written in small letters.
3. The **two names** are underlined when **handwritten** e.g. Ceratotherium simum (white rhinoceros) and Loxodonta africana (African elephant)
4. The names are **italicized** (written in Italics) when **printed** to indicate that the names are specific e.g. e.g. *Panthera leo* (lion) and *Panthera tigris* (tiger).

FIVE KINGDOM CLASSIFICATION

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Organisms are divided into five major kingdoms. R.H Whittaker proposed the five kingdom classification in 1969. The most common system of classification in use today is the Five Kingdom Classification. In this system all living organisms are divided into five kingdoms:

The classification of living organisms according to Whittaker into five kingdoms namely are as follows:

KINGDOM: PROKARYOTAE

Kingdom Prokaryotae - It consists of primitive organisms. The organisms are very small and single celled. They consist of prokaryotes which include species like the bacteria, archae bacteria, cyanobacteria and mycoplasma. Example: bacteria.

KINGDOM: PROTISTA

Protista are single-celled eukaryotes and are mainly aquatic. It includes diatoms, golden algae, euglena and protozoans like amoeba, paramecium, plasmodium etc. They are mostly marine and photosynthetic. Example: paramecium

KINGDOM: FUNGI

Kingdom Fungi or Kingdom Mycota consists of network of thread-like structures called as mycelium. The bodies consist of long, thread-like structures which are called hyphae. These organisms are mostly saprophytes or parasites and also symbionts. This kingdom of fungi also includes lichens, mycorrhiza, etc. Example: aspergillus

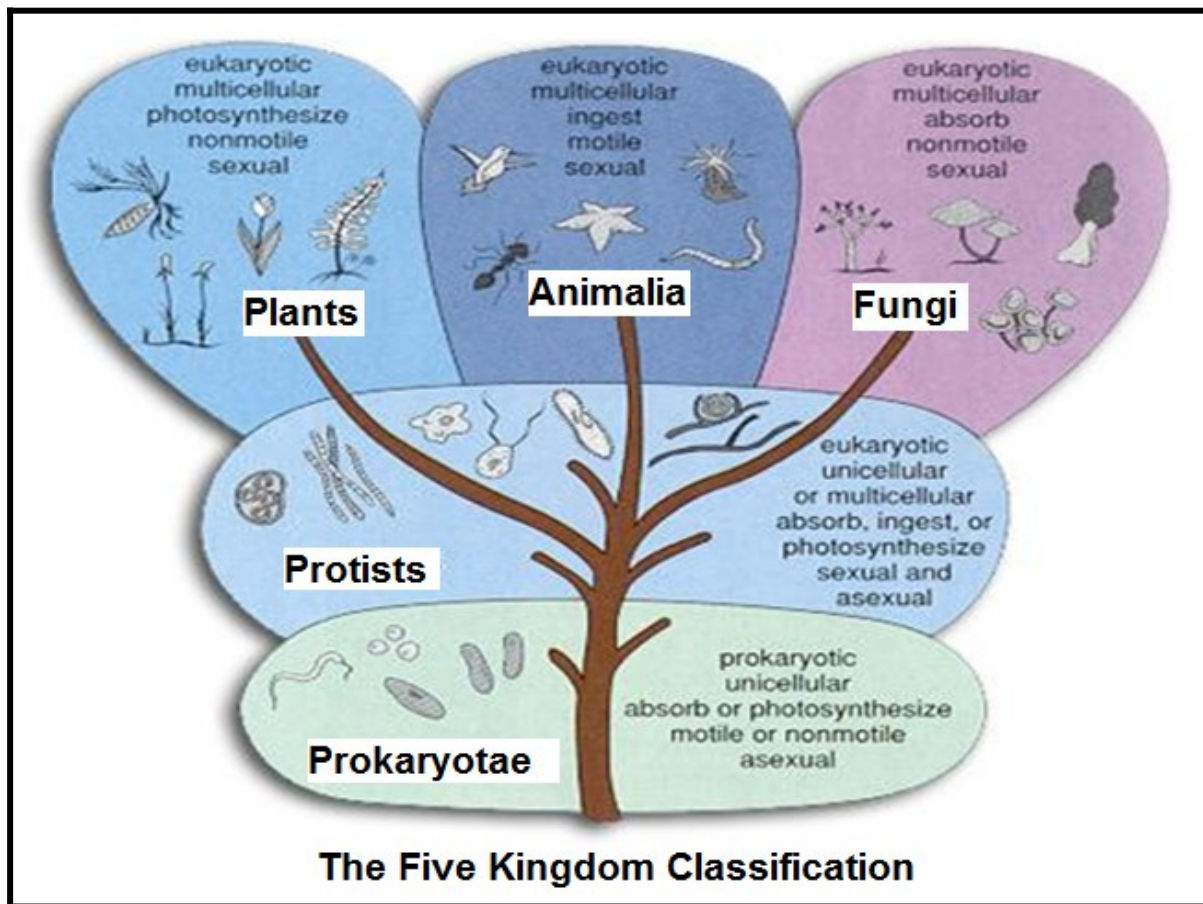
KINGDOM: PLANTAE

Kingdom Plantae or Kingdom Metaphyta are eukaryotic, multicellular plants. They contain chlorophyll pigment, which helps them prepare their own food by the process of photosynthesis. This kingdom includes all types of plants like herbs, shrubs, trees, flowering and non flowering plants. Example: rose plant, mango tree, etc.

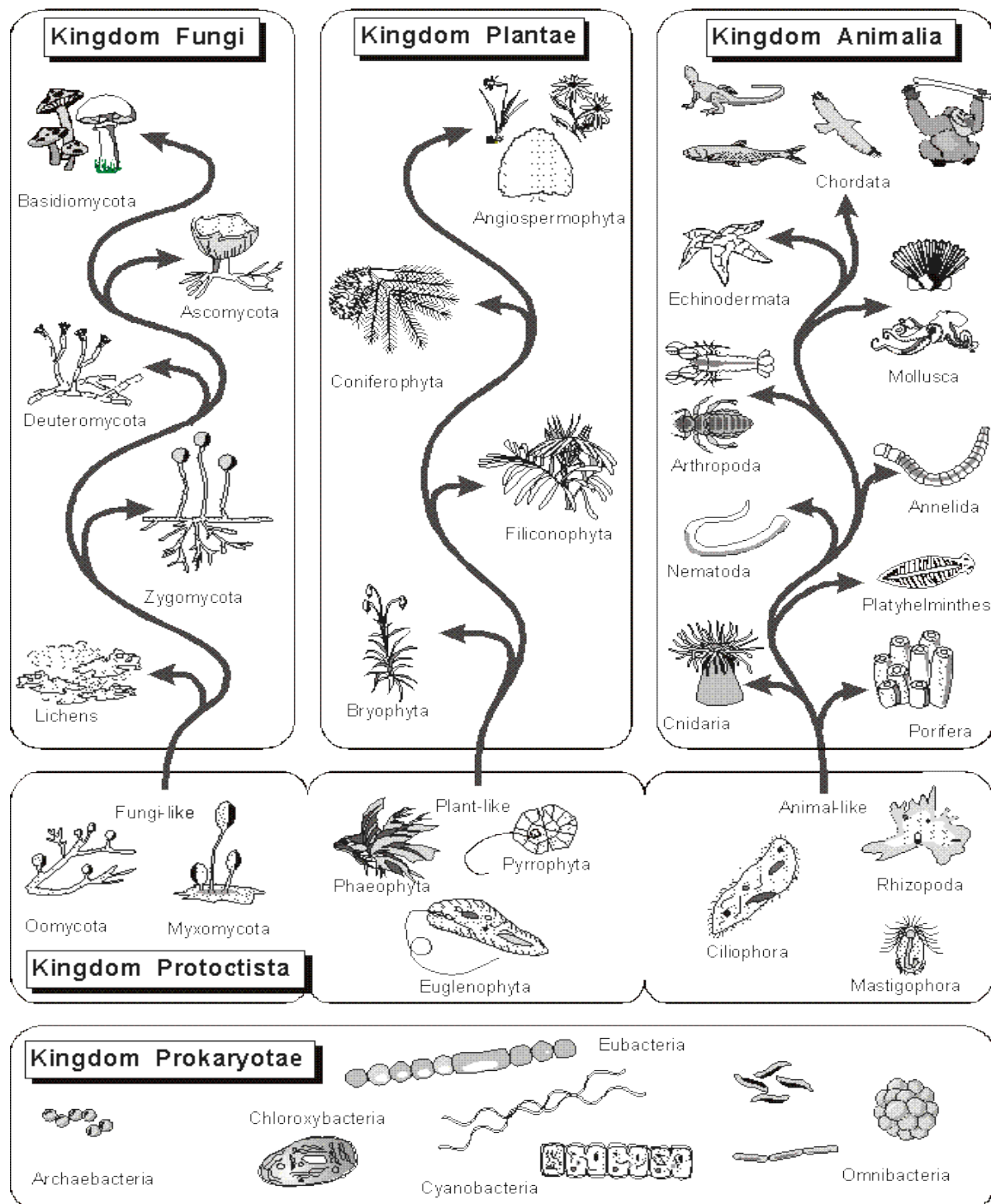
KINGDOM: Animalia

Kingdom Animalia or Kingdom Metazoa are heterotrophic, eukaryotic, multicellular organisms. They lack cell wall. This kingdom includes all types of animals. Example: lion, peacock, etc.

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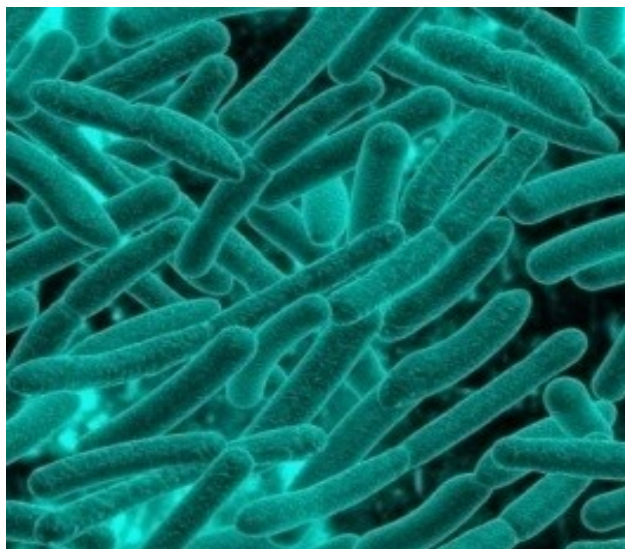
The Five Kingdoms



Kingdom Prokaryotae

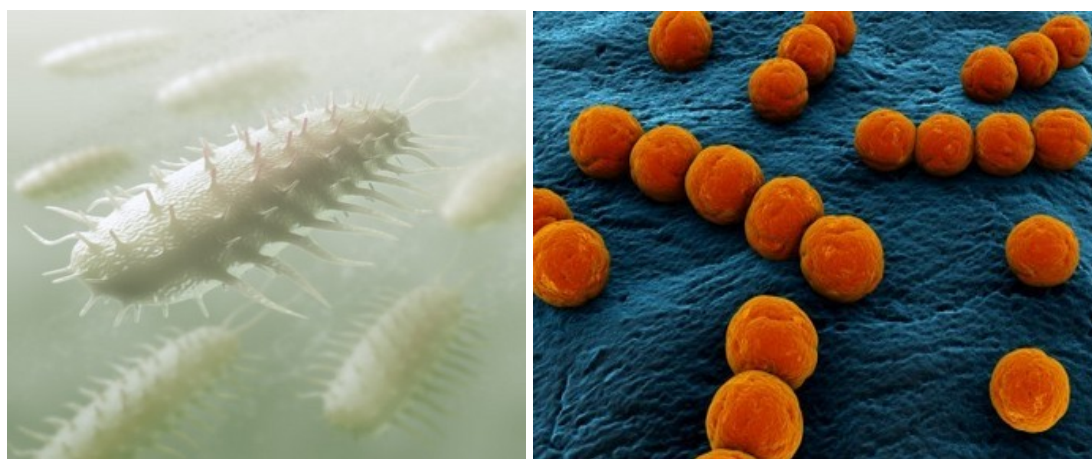
- Kingdom Prokaryotae - All the organisms of this kingdom are prokaryotes. Complex structure was the basis of classification of organisms, many centuries ago. According to R.H. Whittaker's five kingdom classification all the bacteria were placed under the Kingdom Prokaryotae.

- Prokaryotae are considered as the most primitive group of organisms. They include various types of bacteria and blue-green algae.
- Prokaryoteans are most abundant of all organisms, due to their versatility of their habitat. It is estimated that a single drop of water contains 50 billion bacteria.



Definition Kingdom Prokaryotae

The Kingdom Prokaryotae includes organisms that are single-celled known as bacteria. The microorganisms in Kingdom Prokaryotae are considered as the most ancient living forms on earth. The kingdom is divided into two groups *Archaeobacteria* and *Eubacteria*. All the organisms of this kingdom are prokaryotes. These cells do not have nuclear membrane, the chromosome is a single and circular, they also lack membrane bound cellular organelles. This kingdom includes bacteria, cyanobacteria, mycoplasma etc. They are unicellular organisms and do not have specific mode of nutrition. They can be either aerobic or anaerobic. These organisms have cell wall which is made up of peptidoglycans. The cell organelles are not membrane bound. Cell organelles like endoplasmic reticulum, mitochondria are absent. Reproduction is by spore formation and binary fission.



Kingdom Prokaryotae Characteristics

Diagnostic features of the kingdom Prokaryotae

- All organisms of the kingdom are prokaryotes.
- Unicellular
- Lack true nucleus (Membrane bound nucleus is absent).
- Circular D.N.A lies free in the cytoplasm referred to as nucleoid.
- No membrane bound organelles
- Mesosomes for respiration (instead of mitochondria)
- Have 70s ribosomes
- Rigid cell wall of murein (peptidoglycan)
- Average diameter 0.5-5 micrometres
- Reproduce **asexually** by binary fission or **sexually** by conjugation, transformation and transduction.
- Extremophiles (live in harsh conditions).

Other characteristics of kingdom Prokaryotae include the following:

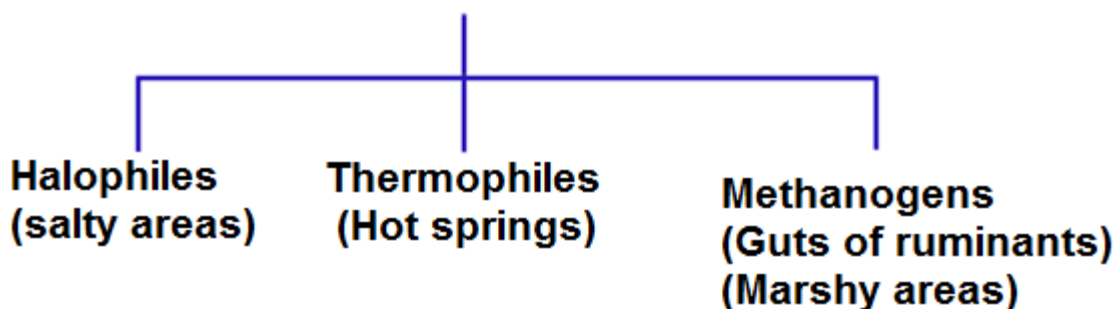
- Habitat - Prokaryotes are found everywhere in hot springs, under ice, in deep ocean floor, in deserts and on or inside the body of plants and animals.
- Nutrition - *autotrophs* - can prepare their own food, *heterotrophs* - depend on others for food, *saprophytes* - feed on dead and decaying matter, *parasitic* - live on other host cells for survival and cause, *symbiotic* - in mutual relation with other organisms, *commensalism* - it is where one organism is benefited and the other is not affected, *mutualism* - where both the organisms are benefited.
- Respiration - respiration in these organisms vary, they may be *obligate aerobes* - the organisms must have oxygen for survival; *obligate anaerobes* - the organisms cannot survive in the presence of oxygen; *facultative anaerobes* - these organisms can survive with or without oxygen.
- Circulation - is through diffusion.
- Movement - is with the help of flagella.

Classification of Kingdom Prokaryotae

Kingdom Prokaryotae has been classified into two groups - Archaeobacteria and Eubacteria.

Archaeobacteria are microbes that live in extreme and harsh conditions, they are known as **extremophiles**. These bacteria lack cell wall, their cell membrane is made up of different lipids, and their ribosomes are similar to that of eukaryotes. Archaeobacteria are of three major groups of bacteria based on their habitat i.e., thermophiles, halophiles and methanogens.

BASED ON HABITAT



Some extremophiles live in boiling water like hot springs and in volcanoes. These are known as '*thermophiles*'.

Some extremophiles live in extremely salty water, the salt loving bacteria are known as '*halophiles*'.

Some bacteria are present in the guts of ruminants and are responsible for production of methane gas from their dung. These bacteria are known as '*methanogens*'.

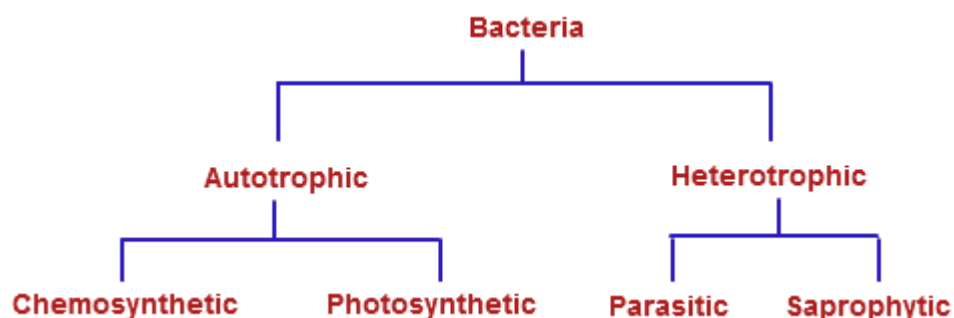
Eubacteria are true bacteria. The characteristic feature is the presence of rigid cell wall and if present a motile flagellum that aids in locomotion. These organisms are characterized based on their nutrition and their shapes.

Classification based on Shape

Bacteria can be classified in four groups based on shape: Spherical or round shaped bacteria are called cocci, Rod-shaped are bacilli, Comma-shaped bacteria are vibrio and spiral shaped bacteria are spirilla.

Classification Based on Mode of Nutrition

Based on the mode of nutrition bacteria are broadly classified into Autotrophic and Heterotrophic.



Autotrophic bacteria - Bacteria which prepare their own food are autotrophic. (Example *Cyanobacteria*)

Heterotrophic bacteria - Bacteria which are dependent on other organisms for their food are heterotrophic. (Example *Escherichia coli*)

Autotrophic bacteria can be Chemosynthetic or Photosynthetic.

Chemosynthetic bacteria are those which prepare their food with the help of inorganic substrates. Photosynthetic bacteria are autotrophic bacteria which prepare their own food by the process of photosynthesis.

Cyanobacteria (Blue-green algae) - They have chlorophyll similar to plants and hence they are photosynthetic autotrophs. The marine and terrestrial and they may be unicellular, colonial or filamentous. The colonies are surrounded by gelatinous sheath. They can also fix atmospheric nitrogen.

Example: *Nostoc* and *Anaebena*.

Chemosynthetic autotrophs - these organisms oxidise substances like nitrites, nitrates, ammonia etc. They help in recycling substances like nitrogen, sulphur, iron etc.

Heterotrophic bacteria are those which are dependent on other organism either directly or indirectly for their nutrition. They are most abundant and are important decomposers. Some are helpful in curdling milk, production of antibiotics, in nitrogen fixation and some are pathogens.

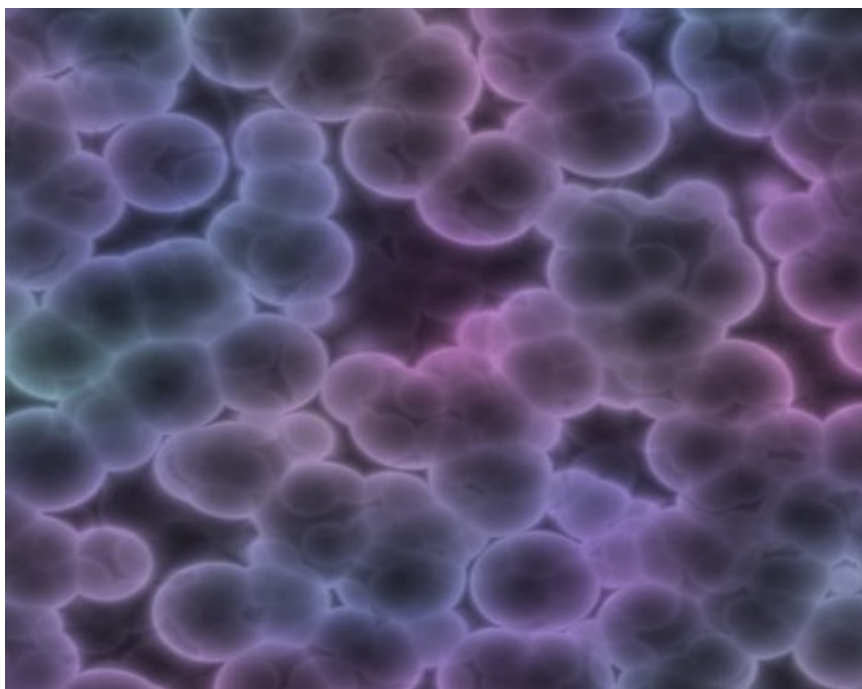
Heterotrophic bacteria can be parasitic and saprophytic. Parasitic bacteria are those which depend on the host for nutrition and cause harm to the host. Saprophytic bacteria feed on dead and decaying matter.

Symbiotic - it is a type where the bacteria are in mutual relation with other organisms. Symbiosis is of two types mutualism and commensalism. Mutualism is where the bacteria and the other organism are benefited due to the relationship. Commensalism is a relationship where the bacteria is benefited while the other organism is not affected by the relationship.

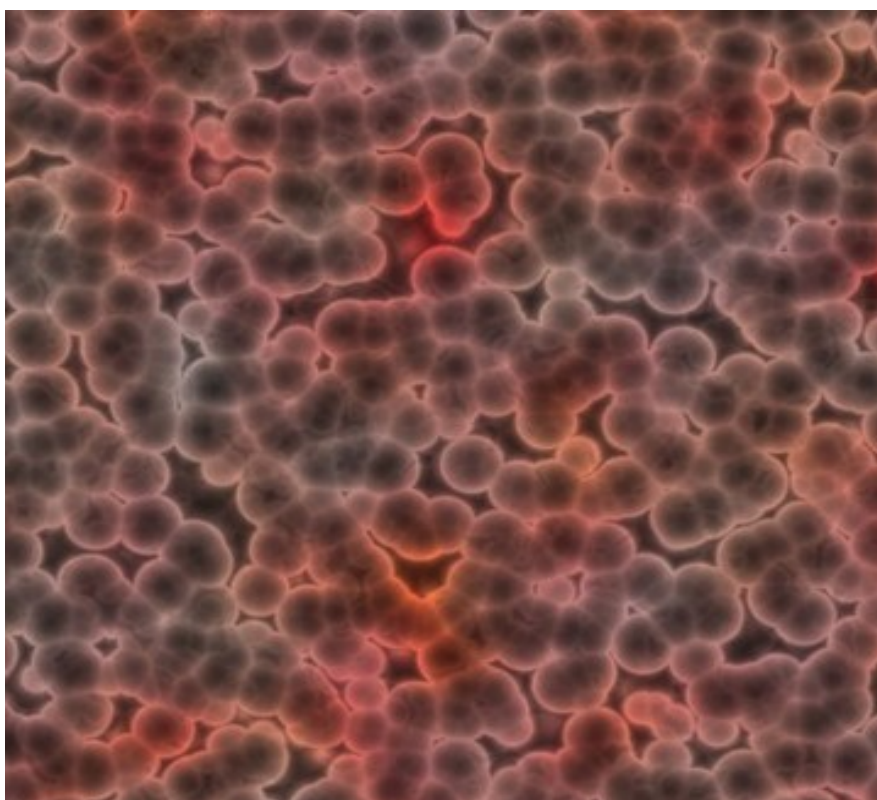
Classification based on Gram's staining

Gram's staining is a test on cell walls developed by Hans Christian Gram. This method helps classifying bacteria into Gram positive bacteria and Gram negative bacteria.

Gram Positive Bacteria - The bacteria's cell wall is made up of protein-sugar complex that takes on purple color during gram staining.



Gram Negative Bacteria - The gram negative bacteria has an extra layer of lipid on the outside of the cell wall and appear pink during the Gram staining procedure.



Reproduction in Bacteria

Reproduction in bacteria is mainly by fission. Under unfavourable conditions they reproduce by spores. Sexually bacteria reproduce by a primitive mode of DNA transfer from one bacterium to another i.e., by conjugation, transduction or transformation.



Mycoplasma

Mycoplasma are known to be the smallest living cells. They completely lack cell wall and can survive without oxygen. Most of the mycoplasma are pathogenic in nature in animals and plants.

SOCIO-ECONOMIC IMPORTANCE OF KINGDOM PROKARYOTAE

- fermentation
- bio-technology
- food spoilage
- decomposition

Economic Importance of Bacteria

1. **Fermentation:** Bacterial fermentation is used in the manufacture of butter, yogurt, cheese, cakes and beer. Bacterial fermentation is also used to produce biogas, an alternative source of energy.
2. **Bio-technology:** *Bacillus thuringensis* bacteria can be bio-engineered for the production of therapeutic proteins like insulin, growth factors and antibodies, etc.
3. **Food spoilage:** Bacteria cause food spoilage and reduce the shelf life of fruits and vegetables.
4. **Decomposition:** Bacteria decompose dead organic matter and wastes.
5. **Bioremediation:** The ability of the bacteria to degrade organic compounds is used in processing toxic wastes and cleaning oil spills.
6. **Nutrient recycling** e.g. nitrifying bacteria in the nitrogen cycle.
7. **Biological control of pests:** In pest control, bacteria can be used in the place of pesticides as these pesticides are regarded environmentally friendly.

Kingdom Prokaryotae Examples

The Prokaryotae comprises of mostly bacteria. Following are a few well known examples.

Blue-green algae - *Cyanobacteria*, Cocci shaped bacteria - *Streptococcus*, Bacilli

shaped bacteria - *E.coli*, Vibrio shaped - *Vibrio cholerae* (cholera), Spiral shaped bacteria - *Treponema pallidum* (syphilis).

Gram positive bacteria - *Mycobacterium*; Gram negative bacteria - *E.coli* (coliforms)

Escherichia coli, *Pseudomonas aeruginosa*, *Proteus vulgaris*, *Streptococcus pyogenes*, *Clostridium botulinum*, *Listeria monocytogenes*, *Salmonella enterica*, *Vibrio cholerae*, *Bacillus anthracis*, *Bacillus cereus*

Kingdom Protista

- Made up of unicellular eukaryotes no longer classified as animals, plants or Fungi e.g. algae and protozoa. The term Protista was first used by Ernst Haeckel in the year 1886
- Members of Protista are primarily aquatic in nature. It is a very large group comprising of at least 16 phyla.
- Many protists like algae are the primary producers in the aquatic ecosystem. some protists e.g. protozoa are responsible for serious human diseases like malaria and sleeping sickness.

Kingdom Protista Definition

Kingdom Protista is a diverse group of eukaryotic organisms. Protists are unicellular, some are colonial or multicellular, they do not have specialized tissue organization. The simple cellular organization distinguishes the protists from other eukaryotes. The cell body of the protists contain a nucleus which is well defined and membrane bound organelles. Some have flagella or cilia for locomotion. Reproduction in protists is both asexual and sexual. They live in any environment that contains water.

Characteristics of Kingdom Protista

General characteristics of Kingdom Protista are as follows:

- They are simple eukaryotic organisms.
- Most of the organisms are unicellular, some are colonial and some are multicellular like algae.
- Most of the protists live in water, some in moist soil or even the body of human and plants.
- These organisms are eukaryotic, since they have a membrane bound nucleus and endomembrane systems.
- They have mitochondria for cellular respiration and some have chloroplasts for photosynthesis.
- Nuclei of protists contain multiple DNA strands, the number of nucleotides are significantly less than complex eukaryotes.
- Movement is often by flagella or cilia.
- Protists are multicellular organisms, they are not a plant, animal or fungus.

- Respiration - cellular respiration is primarily aerobic process, but some living in mud below ponds or in digestive tracts of animals are strict facultative anaerobes.
- Nutrition - they can be both heterotrophic or autotrophic.
- Flagellates are filter feeding, some protists feed by the process of endocytosis (formation of food vacuole by engulfing a bacteria and extending their cell membrane).
- Reproduction - some species have complex life cycle involving multiple organisms. Example: Plasmodium. Some reproduce sexually and others asexually.
- They can reproduce by mitosis and some are capable of meiosis for sexual reproduction.
- They form cysts in adverse conditions.
- Some protists are pathogens of both animals and plants. Example: Plasmodium falciparum causes malaria in humans.
- Protists are major component of plankton.

PROTOZOA	ALGAE
Non-photosynthetic	photosynthetic
Parasitic and some free living	Free living/non parasitic
No cell walls	Have cellulose cell walls
Small and temporary food vacuoles	Large permanent vacuoles
Some motile and some non -Non motile motile	
Unicellular	Multicellular or unicellular
No tissues formed	No tissues

Kingdom Protista Classification

Kingdom Protista are categorized into two taxons:

Protozoans - animal-like single-celled organisms.

Algae - plant-like single or multi-celled organisms.

Animal-like Protists - PROTOZOANS

Protists that have resemblance to animals are known as protozoans. They live in moist and watery environments. The characteristics similar to animals are - their ability to move and their *inability* to produce their own food (heterotrophs). They differ from animals being unicellular while animals are multicellular.

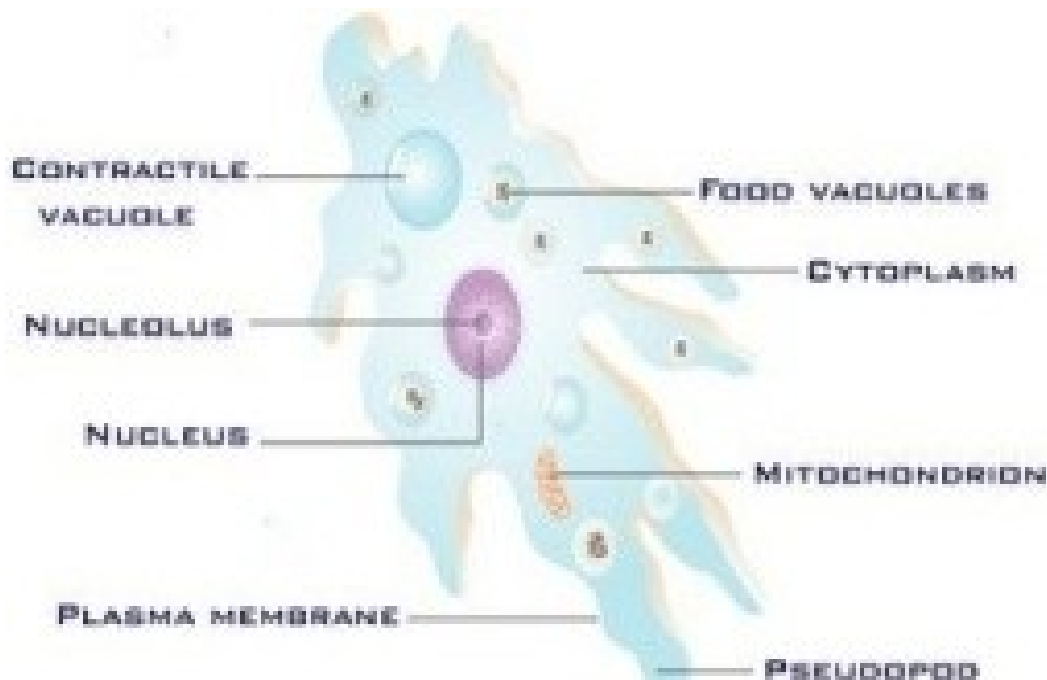
Protozoans are classified on the way they move into four categories:

- *Sarcodinians* - move using pseudopod.
- *Zooflagellates* - move using flagella.
- *Ciliophorans* - move using cilia.
- *Sporozoans* - forms spores.

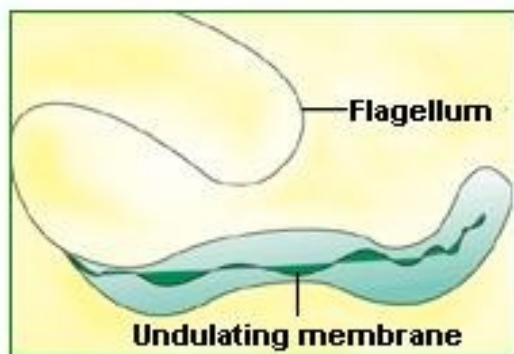
Phylum Sarcodina - The movement in sarcodinians is by extending lobes of cytoplasm known as pseudopodia. The pseudopoda is used for movement and

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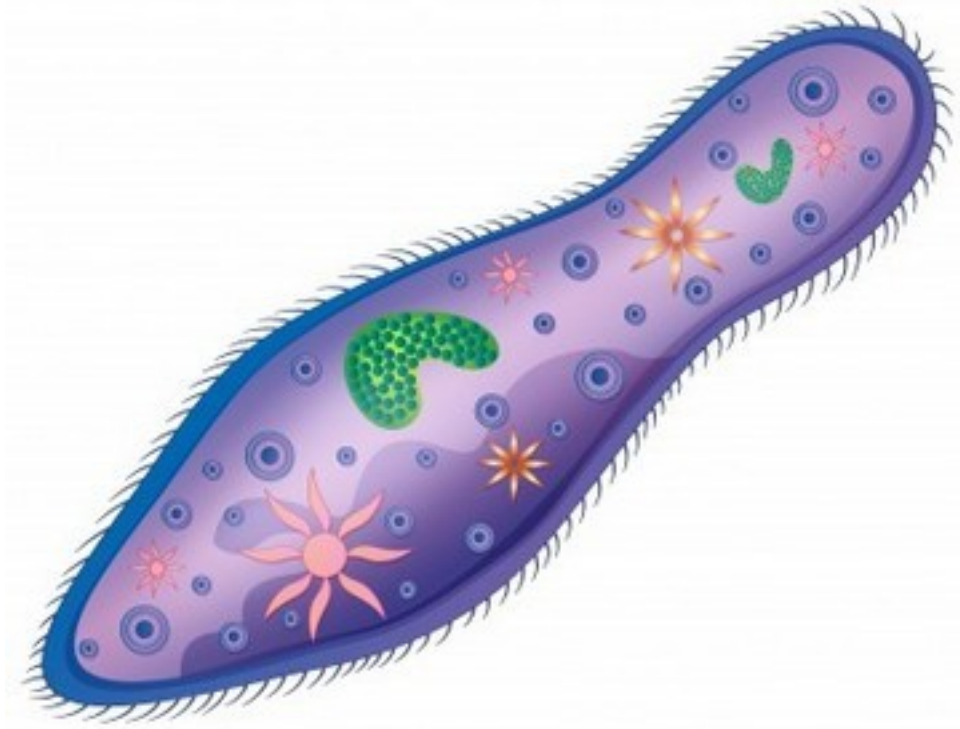
feeding. During the formation of the pseudopodia the cytoplasm streams into the lobe causing the lobe to 'ooze' and grow. Because of this the pseudopodians have a 'blob like appearance'. Example: *Amoeba*, *Foraminiferans*.



Phylum *Mastigophora* (Zooflagellata) - These protozoans move with the help of flagella. Most of them are parasitic. Many flagellates are seen in the intestine of humans, in termites and other animals, some flagellates are harmful. Example: *Trypanosoma gambiense* causes sleeping sickness in cattle and human.



Phylum *Ciliophora* (Ciliates) - Protozoans of this phylum move with hair like structures called cilia. The cilia stick out of their cells. The movement of cilia is paddle like, it sways back and forth for movement and fast beating of the cilia causes movement of the organism. The cilia is also used to sweep food particles into the organism. Example: *Paramecium* - It is a ciliate protozoan found in fresh water and ponds. It commonly known as the slipper animal-cule.



Phylum *Sporozoa* - All members of this phylum are non-motile and parasitic. They form spores and hence the name sporozoa. They lack locomotory structures and they are carried in their hosts by their body fluids. Many sporozoans cause serious diseases in humans. Example: *Plasmodium* - this parasite causes malaria in humans.

Plant-like Protists - ALGAE

Plant-like protists have chlorophyll like that in plants. This chlorophyll enables them to make food by photosynthesis. They produce and release oxygen like the plants. It is believed that the most supply of oxygen on Earth is from the plant-like protists. The plant-like protists are the major food source and primary producers for water organisms.

Phylum *Chlorophyta* (Green Algae) - The green algae include unicellular and multicellular algae. They are mostly fresh water. Body is sheet-like thallus. They have cell walls made of cellulose and pectin. Food is reserve starch which is stored in pyrenoids. Example: *Spirogyra* - it is a unicellular green algae, it grows as a green thread or filament.



Phylum *Rhodophyta* (Red Algae) - Red algae are mostly large and multicellular. They grow in oceans. The algae 'Nori' and Gelidium are used as food, in parts of Asia. Carragean and agar are glue-like substances in red-algae. Agar is used as a medium used for growing bacteria and other organisms under laboratory conditions. Agar is also used to make gelatin capsules and a base for cosmetics. Carragean is used as a stabilizer and thickener in dairy products. It is also used to give toothpaste its creamy texture.



Phylum *Phaeophyta* (Brown Algae) - Brown algae are multicellular. They grow on rocks in shallow water of the sea. Large brown algae are called kelps. Kelps may grow densely in the sea and form kelp forests. They form important food sources for fish and invertebrates. The brown algae growing on rocks are known as rockweed. Example of rockweed is *Sargassum*. Algin is a substance derived from some algae which is used in making ice cream, lotion and plastics.



Phylum *Chrysophyta* (Golden algae) - 'Chryso' means 'color of gold'. There are three types of golden-algae: yellow-green algae, golden brown algae, diatoms. Diatoms are the most abundant and are found in seawater and freshwater habitats. The shell of diatoms are made of silica. They are major source of food to many aquatic organisms. The shells of fossil diatoms form thick deposits on the sea floor known as 'diatomaceous earth'. It is used as water filters, abrasive and to add sparkling to products such as paint and fingernail polish.



Phylum *Pyrrophyta* (Fire Algae) - It contains of species of one-celled algae called dinoflagellate which means 'spinning swimmers'. They store food in the form of starch and oils. The red color is due to chlorophyll a and c and xanthophylls. These organisms have ability of bioluminescence. Almost all species live in marine water. Some species causes the 'red tide phenomenon'. The dinoflagellates causing red

tides are known as *Gonyaulax*, which contain a neurotoxin and are poisonous to marine fauna.



Fungus-like Protists

Slime Molds - Slime molds are saprophytic protists. They are very bright in appearance. They live in moist soil, decaying plants and trees. They are single-celled organisms. During favourable condition they form multicellular aggregations called plasmodium. During unfavourable conditions, plasmodia differentiate to form fruiting bodies bearing spores at the tip. These spores possess resistant true walls, which help in survival for a long time during adverse conditions. These spores disperse by air currents.



Kingdom Protista Examples

Few well known examples are follows:

Green algae - *Spirogyra*, *Ulva*, *Chlamydomonas*, *Volvox*.

Red algae - *Porphyra*, *Rotalgen*.

Brown algae - *Laminaria*, *Nereocystis*.

Water molds - *Saprolegnia*.

Phylum Ciliata - *Paramecium*, *Vorticella*.

Phylum Dinoflagellata - *Ceratium*, *Gonyaulax*.

Phylum Mastigophora - *Trypanosoma*, *Trichonympha*.

Phylum Sarcodina - *Amoeba*.

SOCIO-ECONOMIC IMPORTANCE OF KINGDOM PROTISTA

- Plasmodium sp - malaria
- *Schistosoma* sp - Schistosomiasis
- Trypanosoma sp - Trypanosomiasis

Economic Importance of Protists

1. **Pathogens:** Some protists are **harmful** as they cause human diseases e.g.
 - the **Plasmodium** species cause **malaria**,
 - the **Schistosoma** species causes **Schistosomiasis (bilharzia)**
 - the **Trypanosoma** causes **Trypanosomiasis (sleeping sickness)**.
2. **Pathogens:** Some protists are **harmful** as they cause plant diseases e.g. late potato blight.
3. The Protists are **useful** in the following manners:
 - (a). **Source of food** - Some protists like kelps are edible.
 - (b). **Source of commercial products** - Marine protists are sources of useful substances like algin, agar, carrageenan and antiseptics.
 - (c). **Primary producers** of aquatic ecosystem - Many protists are primary producers; they play a basic role in food chains, providing food and oxygen.
 - (d). **Source of medicines** - Sodium laminaria sulphate, Fucoidin, Heparin are algal products used as blood coagulants. Lyngbya produces an anti-cancer compound.
 - (e). **Source of minerals** - Kelps are rich in sodium, potassium, iodine etc, they are good source of minerals.
 - (f). **Biological research** - They are used in biological research, e.g., Chlorella is unicellular, non-motile alga.

Kingdom Fungi

The organisms in kingdom fungi include mushrooms, yeasts, molds, rusts, smuts, puffballs, truffles, morels, and molds. More than 70,000 species of fungi have been identified. The fungi constitute an independent group to that of plants and animals. They live everywhere in air, in water, on land, in soil, and on or in plants and animals. Some fungi are microscopic and others extend for more than a thousand acres. Mycology is a discipline of biology which deals with the study of fungi. Fungi appear like plants but are closely related to animals. Fungi are not capable of producing their own food, so they get their nourishment from other sources. Fungi are in a wide variety of sizes and forms and have great economic importance.



Kingdom Fungi Definition

Fungi show a great diversity in morphology and habitat. Fungi are heterotrophic organisms, they obtain their nutrients by absorption. The cell wall of fungi are mostly made up of the carbohydrate chitin, while the cell wall in plants is made of cellulose. The carbohydrates stored in fungi is in the form of glycogen. The 'fruit' body of fungus is only seen, while the living body of the fungus is a mycelium, it is made of tiny filaments called hyphae. The mycelium is hidden. Nutrition in fungi is by absorbing nutrients from the organic material in which they live. Fungi do not have stomachs, they digest their food externally before it pass through the cell wall into the hyphae. The hyphae secretes enzymes and acids that break down the organic material into simple compounds.

Kingdom Fungi Characteristics

Diagnostic features of the Kingdom Fungi

- eukaryotic
- some are unicellular e.g. yeast and some are multicellular e.g. mushroom
- non photosynthetic store carbohydrate as glycogen
- asexual reproduction by spore formation
- non-motile
- heterotrophic/saprotrophic/parasitic/mutualistic
 - (**Heterotrophic** nutrition: use organic compounds (carbon) made by other organisms as its source of energy and molecules for metabolism from:
 - ❖ dead and decaying matter
 - ❖ feeding as parasites on living organisms).
 - Nutrition in fungi - they are saprophytes, or parasites or symbionts.

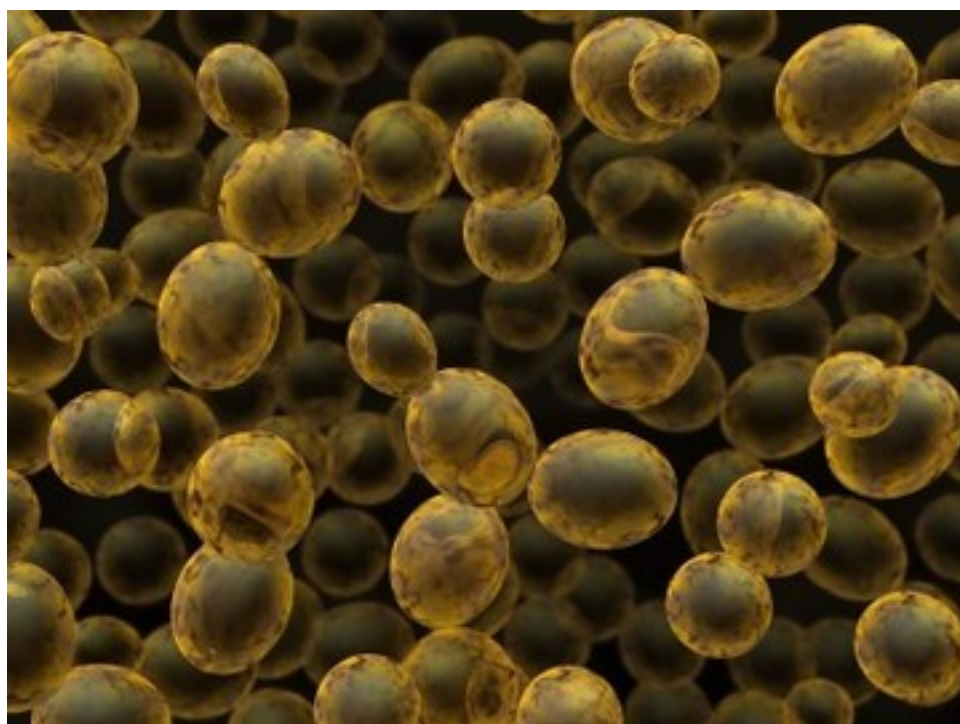
- absorptive-digestion. It takes place outside the body and nutrients are absorbed. The fungi digest the food externally first and then ingest the food, to accomplish this the fungi produce exoenzymes.
- cell walls made of chitin.
- body is a mycelium a network of fine tubular filaments called hyphae growing from horizontal hyphae the stolon
- end of hyphae bears sporangia which are a reproductive organ for spore formation
- Fungi exhibit the phenomenon of **alteration of generation**.

Kingdom Fungi Classification

Based on the spore case in which the spores are produced fungi are classified into four divisions.

Division Ascomycota: Sac Fungi

The sac-fungi produce spores in small cup-shaped sacs called asci, hence the name ascomycota. The mature sac fungi spores are known as ascospores, they are released at the tip of the ascus breaks open. Yeast is the most common one-celled fungi. Yeast reproduces through asexual process called budding. The buds form at the side of the parent cell, they pinch-off and grow into new yeast cell which is identical to the parent cell. Examples of sac-fungi are morels, truffles, cup fungi and powdery mildews. Example: *Aspergillus*, *Claviceps*, *Neurospora*.



Division Basidiomycota: Club Fungi

Basidiomycota includes the mushrooms, puff-balls, smuts, rusts and toadstools. The spores are borne on a club-shaped spore case called basidium. In mushrooms the basidia are lined at the gills under the cap. Huge numbers of spores are produced by the club fungi. In fact, an average sized mushroom produces over 16

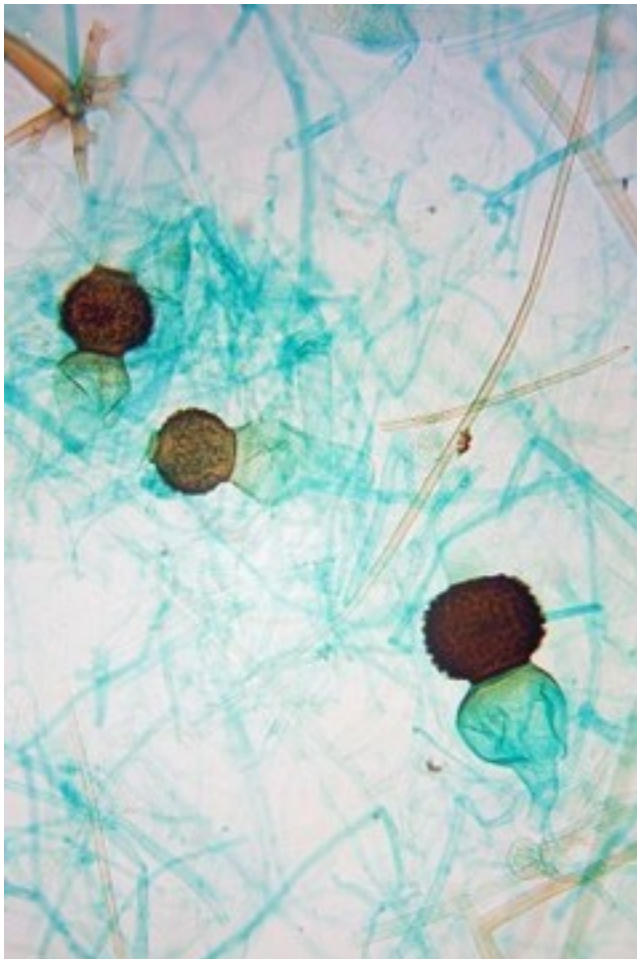
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billion spores. These spores rarely germinate or mature. Example: *Agaricus*(mushroom), *Ustilago*(smut), and *Puccinia*(rust fungus).



Division Zygomycota: Zygote forming Fungi

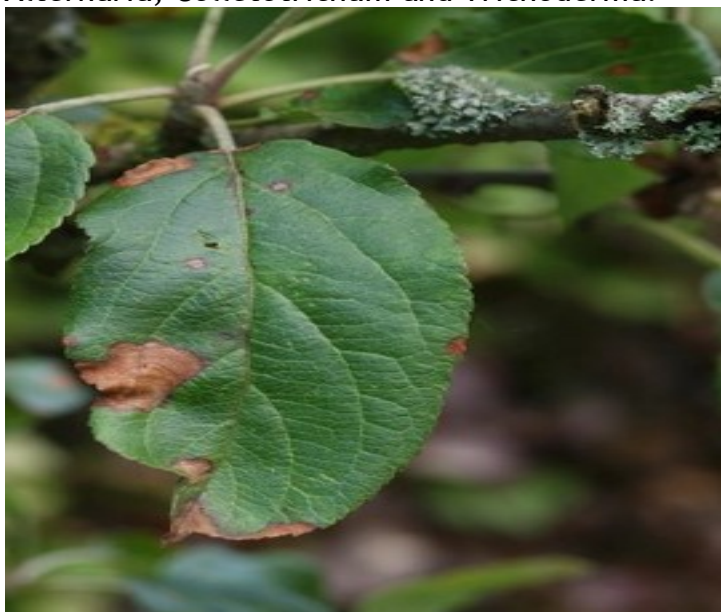
These fungi are usually found on cheese, bread, and other decaying food. They are zygote forming fungi, hence the name zygomycota. The spores are produced in round-shaped case called sporangium. The grayish fuzz seen on bread and decaying food is actually mass of mature sporangia mold. Under the microscope they are seen as pinheads. When the sporangium breaks open hundreds of spores are released. Example: *Mucor*, *Rhizopus* (the bread mould) and *Albugo*.



Division Deuteromycota: Imperfect Fungi

These organisms are known as imperfect fungi because they lack sexual reproduction. They reproduce by asexual spores known as conidia. Most of the fungi causes diseases to humans like ringworm, athlete's foot. Economically

important imperfect fungi are *Penicillium* and *Aspergillus*. Other examples are *Alternaria*, *Colletotrichum* and *Trichoderma*.



Kingdom Fungi Examples

Some of the examples of kingdom fungi are as follows:

Sac-fungi : *Agaricus* (mushroom), *Ustilago* (smut), and *Puccinia* (rust fungus).

Zygote-forming fungi : *Mucor*, *Rhizopus* (the bread mould) and *Albugo*.

Club fungi: *Agaricus* (mushroom), *Ustilago* (smut), and *Puccinia* (rust fungus).

Imperfect fungi: *Alternaria*, *Colletotrichum* and *Trichoderma*.

Members of the Kingdom Fungi

Mycorrhizae - More than 90% of the plants are symbionts of mycorrhizae. Myco means fungus and rhiza means root. Mycorrhizae are of two types ectomycorrhizae and endomycorrhizae.

Ectomycorrhizae - These are fungus forms sheath outside the root.

Endomycorrhizae - They are also known as vesicular-arbuscular-mycorrhizae (VAM). Fungus does not form sheath around the roots.

Lichens - They are symbionts. They have a symbiotic relationship between a fungus and an alga. Neither of the organisms can survive on their own.

SOCIO-ECONOMIC IMPORTANCE OF KINGDOM FUNGI

Can be positive (beneficial) or negative (harmful/detrimental):

1. Beneficial (Positive) effects of bacteria

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- (a). **Decomposition/nutrient recycling:** increase soil fertility by decomposing dead matter. e.g. saprophytic fungi
- (b). **Decomposition/sewage treatment:** Some fungi decompose sewage.
- (c). **Fermentation:** used in brewing industry and baking industry, e.g. yeast
- (d). **Penicillin production:** antibiotics formation e.g. Penicillium produce useful penicillin
- (e). certain fungi are particularly suitable for use in the study of genetics
- (f). **Food:** act as food sources, e.g. mushrooms are rich in protein
- (g). **Bio-control Agents:** Fungi are used to parasitise insects which help control pests. Spores of fungi are sprayed on crops, this method is cheaper and environmentally friendly.
- (h). **Mycorrhizae** are fungi associated with roots of plants. Roots benefit in getting minerals from the environment while fungi get food from the plant in return through such association.

2. Harmful (Negative) effects of bacteria

- (a). **Pathogens:** causing human diseases, e.g. ringworm, athlete's foot
- (b). **Pathogens:** causing plant diseases and reducing yields, e.g. wheat rust.
- (c). **Food spoilage:** cause food spoilage, e.g. Mucor and Rhizopus (bread mould).
- (d). **food poisoning,** e.g. poisonous mushrooms
- (e). **Decomposition:** deterioration of wooden buildings.

Kingdom Plantae

There are different types of plant species, which are found on planet earth. They are sorted and classified into a separate kingdom known as Kingdom plantae. This classification is based on their similarities and differences.

The Kingdom plantae is also called as kingdom Metaphyta. The Kingdom plantae includes all types of eukaryotic, multicellular, photosynthetic plants found in this biosphere. Most of the organism in this kingdom is autotrophs, which synthesis their own food with the help of solar energy. There are very few species, which are both autotrophs and heterotrophs. The history of life on earth and the success of many organisms literally depend on the success of plants.

Kingdom Plantae



Kingdom Plantae Definition

The Kingdom plantae can be defined as multicellular, autotrophic eukaryotes, which conduct photosynthesis. All member of this family comprises of true nucleus and advanced membrane bound organelles. They are quite different from animals. The Kingdom Plantae contains about 300,000 different species of plants. Among the five kingdoms, Kingdom plantae is a very important, as they are the source of food for all other living creatures present on planet earth, which depends on plants to survive.

Kingdom Plantae Characteristics

Below you could see characteristics for kingdom plantae

- Most of the plants are eukaryotic and chlorophyll containing organisms.
- Cell walls of plant cells are comprised of cellulose.
- They have an ability to grow by cell division.
- In life cycle of plant cells, the interchanges occur from the embryos and are supported by other tissues and self produce.
- Plants have both organs and organ systems.
- They obtain their energy from sun through photosynthesis.
- Plants reproduce both by sexual and asexual.

- Plants develop a self defence mechanism to protect them from being destroyed by animals, fungi and other plants.
- Organisms within Kingdom Plantae are multicellular, eukaryotic and autotrophic.
- They lack motility.

Examples of kingdom Plantae

This kingdom includes all types plants like *Zea mays*, herbs, shrubs, trees, creepers, climbers, aquatic plants, desert plants, mountain plants, flowering and non flowering plants, etc.





Kingdom Plantae Classification

The Kingdom Plantae contains about billion types of plants species and it was very difficult to identify different types of species. Many biologists contributed in classifying different plants species in to their separate kingdom.

Based on their classification, plants are divided into the four main groups. These classifications was based on

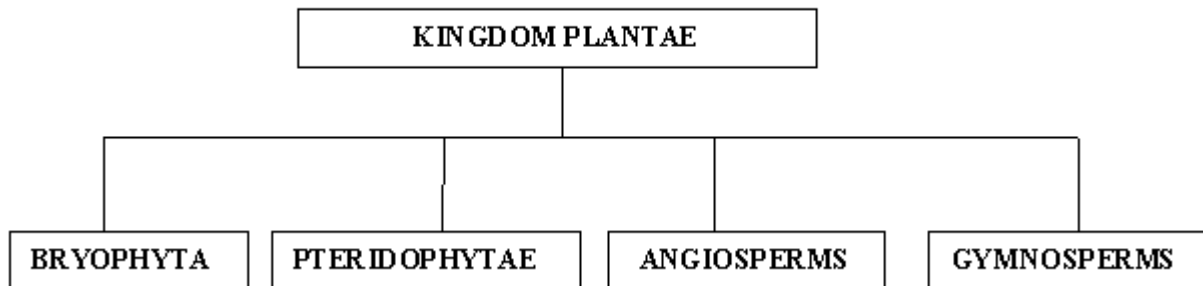
- The presence of vascular tissue.
- The absence of vascular tissue.
- The presence of seeds.
- The absence of seeds.

Vascular tissue and the seeds are mainly considered as they play a vital role in:

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Vascular tissue helps in transporting substances (water, minerals and sugars) throughout the plant.

Seeds are structures, which contains an embryo that helps in storing food.



Phylum Bryophyta

- They are non-vascular land plants, which do not contain any conducting tissues and are often referred to as bryophytes.
- These plants are small, grow close to the ground and include mosses and liverworts.
- They are very small in structure and are considered as important members of our ecosystem.
- The reproduction process is carried in their spores. They are non flowering plant and are found mainly growing on the ground, on other plants and on rocks.
- They play a vital role in preventing soil erosion.

Phylum bryophyta e.g.: Mosses, Liverworts and Hornworts.

Mosses



Phylum Pteridophytae

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- They are seedless vascular plants, which contain vascular tissues but do not produce seeds.
- They are involved in transportation of fluids.
- The reproduction process is carried by spores.

Phylum pteridophytae e.g.: horsetails, ferns and club mosses.

Horsetails



Phylum Angiosperms

- Angiosperms are flowering plants, which develop the seeds within a protective structure.
- The reproduction process is carried by Angiosperm.
- They develop their seeds within an ovary, which itself is embedded in a flower. After the stage of fertilization, the flower falls and the ovary bulges to become a fruit.
- Angiosperms in the class Dicotyledoneae grow into two seed-leaves (cotyledons).
- An angiosperm's leaf consists of a single, branched, main vein, which originates from the base of the leaf blade. In few plants, it may also consist of four or more main veins diverging from the same base.

Phylum angiosperms e.g.: trees, shrubs, vines and flowers.

Vines



Phylum Gymnosperms

- Gymnosperms are non-flowering plants with undeveloped seeds, which are present in an enclosed structure.
- Monocot begins with a single seed-leaf. The main veins of their leaves are usually parallel and unbranched.
- Monocot plays an important role in providing us with our primary sources of nutrition, which includes grains, fruits, etc.

Phylum gymnosperms e.g: palms, carpet lawns, etc.

Palms



Non Flowering Plants

There are billion types of plant species found in this biosphere. After the classification of different plants species in to their separate kingdom, biologists again sub classified these plant species into two categories. This classification was based on the plants that produce flowers and their type of reproduction.

Definition of Non Flowering Plants

Non flowering plants can be defined as those plants, which cannot produce flowers and most of them reproduce through spores. They are completely different from flowering plants. These types of plants cannot produce flowers and seeds. Non flowering plants have vascular tissues and they reproduce by spores.

List of Non Flowering Plants

There are more than 700 types of non-flowering plants present on this biosphere, among them some famous non flowering plants are ferns and horsetails. Other examples of non flowering plants are Moses. Liverworts, Cut-leaved Grape Fern, Royal Fern, Cinnamon Fern, Curly-grass Fern, Bead Fern, Lichens, Conifers, Cycads, Algae, Rosidae, Magnoliidae, Gingko, Hornworts, Whisk ferns, Club mosses, etc.

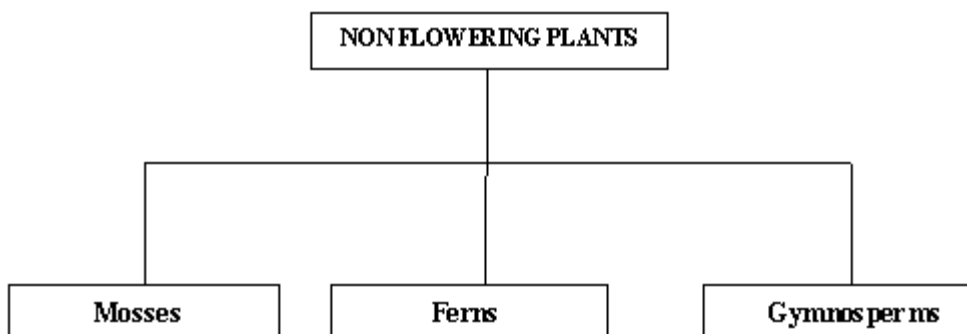
Lichens



Types of Non Flowering Plants

There are three types of non flowering plants. These classifications was based on

- The presence of roots.
- The absence of roots.
- Based on their seeds - spore bearing and naked seeds.



Mosses

General Characteristics of Mosses

They belong to the kingdom of Phylum bryophyta.

- They are the simplest plants with the absence of true roots and vascular tissues.
- They have simple stems, leaves and they do not produce flowers, fruits and seeds.
- They reproduce through spores.
- They are small green coloured plants, which prepares their own food.

- They live in damp shady places.

Mosses



Ferns

General Characteristics of Ferns

They belong to the kingdom of Phylum pteridophytae.

- They are the simplest plants with the presence of roots, feathery leaves and underground stems.
- They have vascular tissues, which helps in the transportation of water, minerals and sugars throughout the plant.
- They have spore producing organs, which is present at the bottom of the leaves.
- They live in damp shady places.
- There are approximately 12,000 varieties of ferns around the globe.
- They are large green coloured plants, which prepares their own food.
- They reproduce through spores.

Ferns



Gymnosperms

- They are tall evergreen trees with the presence of roots, woody stems and needle shaped leaves.
- They have vascular tissues, which helps in the transportation of water, minerals and sugars throughout the plant.
- They contain naked seeds in female cones.
- They reproduce seeds from cones in non-flowering plants.
- They are cone-bearing plants, which resembles to palm trees.
- These trees grow up to 20 feet tall and are found in Mediterranean climates.

Gymnosperms e.g.: hemlock, pine and redwood trees.

Pine tree



Characteristics of Non Flowering Plants

Below you could see non flowering plants characteristics

- These plants do not produce flowers, fruits and seeds.
- Non flowering plants vary in their shape, size and colour.
- They are the first and oldest (about 450 million years ago) plant present on this biosphere.
- They are mainly characterized by spore production, absence of flowers and true roots.
- These plants are simpler compared to those flowering plants.
- They have higher level of adaptability to the environment.
- Reproduction in non flowering plants is asexually by producing spores.
- Gymnosperms are the first non-flowering plants to produce seeds.
Gymnosperms started to produce flowers and fruits after 130 million years ago.

Flowering Plants

A flower is the reproductive organ of all flowering plants. A single flower produces egg and sperm and the entire process of fertilization in plants occurs inside the flower. A single flower gives rise to fruit and seeds. In addition to reproduction flower plays a vital role in pollination by attracting animals, birds and other flies to transfer the pollen grains. According to the history, flowering plants came into existence about 135 million years ago. In fact, the food, which we consume every day comes from the flowering plants itself.



Definition of Flowering Plants

Flowering plants can be defined as those plants, which can produce flowers, fruits and seeds. They are completely different from non-flowering plants. They are also called as angiosperms. Flowering plants are the largest groups within the plant kingdom. There are around 260,000 species of flowering plant and about 90% species are identified. This species also includes aquatic plants, desert plants, land plants, seasonal plants, etc.

Rose - A **flowering plant**.



Examples of Flowering Plants

Sunflower, tulips, marigold, lily, jasmine, rose, lotus, hibiscus, Petunia, daisy, water lilies, Orchid, etc.

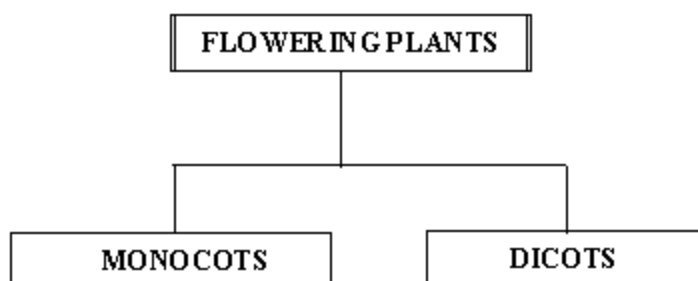
Lotus - An aquatic flowering plant.



Types of Flowering Plants

There are two types of flowering plants. These classifications was based on

- The physical structure of roots, stem and leaves.
- Types of seeds - Monocot and Dicot.



MONOCOTS

General characteristics of monocots.

- They are one seeded plants.
- The leaves of monocotyledons have parallel veins.
- They are herbaceous plants.

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- The parts of the flowers of monocotyledons are arranged in threes or more. It may contain flowers with three petals, flowers with six petals and the stamens also follow this pattern.
- Monocotyledons make a seed with a seed coat.

Monocot e.g. *Zea mays* (corn/maize), grass, rice, wheat, etc.

Corn



DICOTS

General characteristics of dicot.

- They are two seeded plant.
- The leaves of dicotyledons have veins in network.
- Dicotyledons seeds also contain an embryonic plant.
- The flowers of dicotyledons have petals and other parts of flower are arranged in four or five or six. It may contain flowers with four petals, flowers with five petals, flowers have six petals and the stamens also follow this pattern.
- The seed is protected by a seed coat.

Dicot e.g. Bean plant, trees, sunflower, rose, etc.

Sunflower



Characteristics of Flowering Plants

These plants produce flowers, fruits and seeds.

- They are characterized by a root system and a shoot system.
- They are the main producers.
- Reproduction in flowering plants is by sexually.
- Flowering plants make up nearly 90 percent of all plant species found in this biosphere.
- They are present from 130 million years on this biosphere.
- The three largest families of flowering plants are the sunflower, orchid and pea families.

SOCIO-ECONOMIC IMPORTANCE OF KINGDOM PLANTAE

Can be positive (beneficial) or negative (harmful/detrimental):

1. Beneficial

- **Producers** of food for consumers by the process of photosynthesis.
- **Drinks:** natural fruit juices
- **Producers** of oxygen used by animals during respiration.
- **Carbon sink:** absorb CO₂ from air thereby reducing the greenhouse effect/global warming.
- **Timber** used for construction and furniture.
- **Medicinal use** e.g. herbs
- **Cosmetics** e.g. Aloe vera and lavender oil.
- **Tourism:** Eco-tourism generates forex and jobs.
- **Clothing** fibres e.g. cotton.
- **Decoration** of businesses and parks e.g. flowers/ornamental plants.
- **Habitats:** Plants are habitats for many animals.
- **Transpiration** release water vapour for the water cycle.
- **Firewood and charcoal:** fuels for cooking and heating.

- ## 2. Harmful

- # Kingdom Animalia

There are around 9 to 10 million species of animals, and about 800,000 species are identified. Fossil records of animals were found in the era of the Cambrian explosion, about 540 million years ago. Animals are divided into various sub-groups, biologists have identified about 36 phyla within the animal kingdom including birds, mammals, reptiles, fish, amphibians etc.



The word 'animal' is derived from the Latin word *animalis* which means 'having breath'. The Kingdom Animalia is characterized by eukaryotic and heterotrophic organisms. They are multicellular and lack cell wall. They depend directly or indirectly of plants for their food. Food is ingested and digested in their internal cavity and food reserves are stored as glycogen or fat. Nutrition is holozoic, i.e.,

by ingestion of food. Animals follow a definite growth pattern, the adults have a definite shape and size. Higher forms of animals exhibit well developed sensory and neuromotor mechanism. Most of the organisms are capable of locomotion. Reproduction is by copulation of male and female which is followed by development in embryonic stages.

Kingdom Animalia Characteristics

Diagnostic features of the kingdom Animalia

- Eukaryotic
- Multicellular
- Non photosynthetic
- Heterotrophic
- No cellulose cell walls
- Store carbohydrate as glycogen
- No chlorophyll
- Motile
- Have a nervous system (C.N.S)
- Have endocrine system for homeostasis
- Reproduce sexually or asexually
- Body divided into head, abdomen and limbs
- All have an alimentary canal
- Have muscles and skeletal systems for maintaining body shape, protection and to provide support for inner structures
- Have a circulatory system with a transport medium usually blood pumped around the body in vessels by the heart
- Bilateral symmetry except cnidarians and echinoderms
- Triploblastic except cnidarians
- Some are segmented e.g. annelids and arthropods

Kingdom Animalia Classification

Kingdom Animalia has approximately 36 sub-divisions known as 'phyla'. Each phyla share particular properties structurally and functionally which together separate it from other phyla. Below are the most common phyla classified under traditional biological methodology.

Phylum Porifera - They are primitive organisms, most of them are salt-water sponges. They do not have organs or nerve cells or muscle cells. Approximately, 8,000 species exist today. Example: *Sycon*, *Euspongia*, *Spongilla*.

Phylum Coelentrata (Cnidaria) - This group is composed of jelly-fish and other lower aquatic animals. Approximately, 15,000 species exist today. Example: *Aurelia*, *Adamsia*.

Phylum Platyhelminthes - This group consists of flat worms. They inhabit both marine and fresh water habitats and they are mostly endoparasites found in animals. Example: *Taenia*, *Fascicola*.

Phylum Aschelmeinthos - It is a group of round worms, most of them are parasites. This phylum consists of about 80,000 parasitic worms.

Phylum Annelida - They are present in aquatic, terrestrial and are free-living or parasitic in nature. This phylum comprises of segmented worms. Example: Earthworm, Leech etc.

Phylum Arthropoda - This is the largest phylum which consists of insects. There are over 1 million species of insects existing today. Example: Locusts, Butterfly, Scorpion, Prawn.

Phylum Mollusca - It is the second largest phylum. They are terrestrial and aquatic. Example: *Pila*, *Octopus*.

Phylum Echinodermata - This consists of sea stars and sea urchins. There are about 6,000 species. Example: *Asteria*, *Ophiura*.

Phylum Chordata - Animals of this phylum have a characteristic feature of presence of notochord, a dorsal hollow nerve cord and paired pharyngeal gill slits. Within this phylum advanced group called vertebrates which include fish, amphibians, reptiles, birds and mammals.

Kingdom Animalia Examples

Some of the well known examples of Kingdom Animalia

Phylum Porifera - *Sycon*, *Euspongia*, *Spongilla*.

Phylum Coelentrata - *Aurelia*, *Adamsia*.

Phylum Platyhelminthes - *Taenia* (Tapeworm), *Fascicola* (Liver fluke).

Phylum Aschelminthes - *Ascaris* (Round worm), *Wuchereria* (Filaria worm), *Ancylostoma* (hook worm).

Phylum Annelida - *Nereis*, *Pheretima* (earthworm), *Hirudinaria* (blood sucking leech).

Phylum Arthropoda - *Apis* (honey bee), *Bombyx* (Silkworm), *Laccifer* (lac insect), *Anopheles*, *Culex* and *Aedes* (mosquitoes), *Locusta* (locust), *Limulus* (king crab).

Phylum Mollusca - *Pila* (Apple snail), *Octopus* (devil fish), *Pinctada* (pearl oyster), *Aplysia* (sea-hare), *Sepia* (cuttle-fish), *Dentalium* (Tusk Shell), *Chaetopleura* (Chiton).

Phylum Echinodermata - *Ascarias* (star fish), *Echinus* (sea urchin), *Antedon* (Sea lily), *Cucumaria* (sea cucumber), *Ophiura* (brittle star).

Phylum Chordata - *Balanoglossus*, *Ascidia*. All fishes, amphibians, reptiles, birds and mammals.

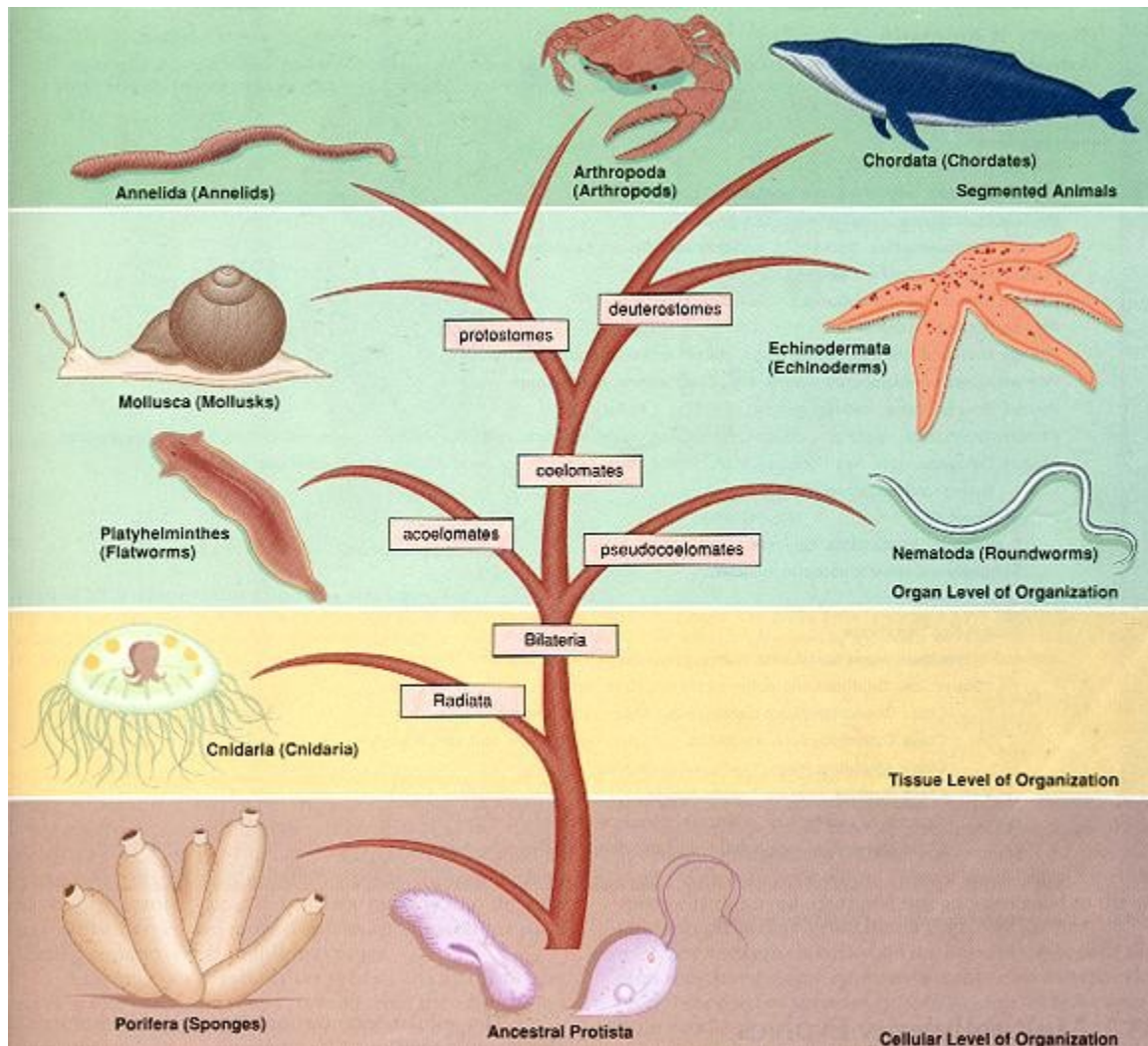
Animal Kingdom Classification

Animals are characterized by being multicellular and eukaryotic. Classification of animals is about organizing organisms into groups. Members of a group have shared characteristic, that is common to all members of that group and it is this character that defines the group.

There are about 1.2 million species of animals identified and there are almost 9-10 million species of animals on earth. This huge population of animals creates

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importance for the need of classification. Classification helps in assigning a systematic position to newly described species.



Characteristics of Animal Kingdom Classification

Basis of Classification

There is a difference in structure and form of different animals, there are a few fundamental characteristics that are common to various organisms. The features are

- Arrangement of cells,
- Body symmetry,
- Nature of coelom,
- Patterns of digestive, circulatory and reproductive systems,
- Arrangement of cells in germ layers,
- Segmentation,
- Notochord. These are the features that forms basis for animal classification.

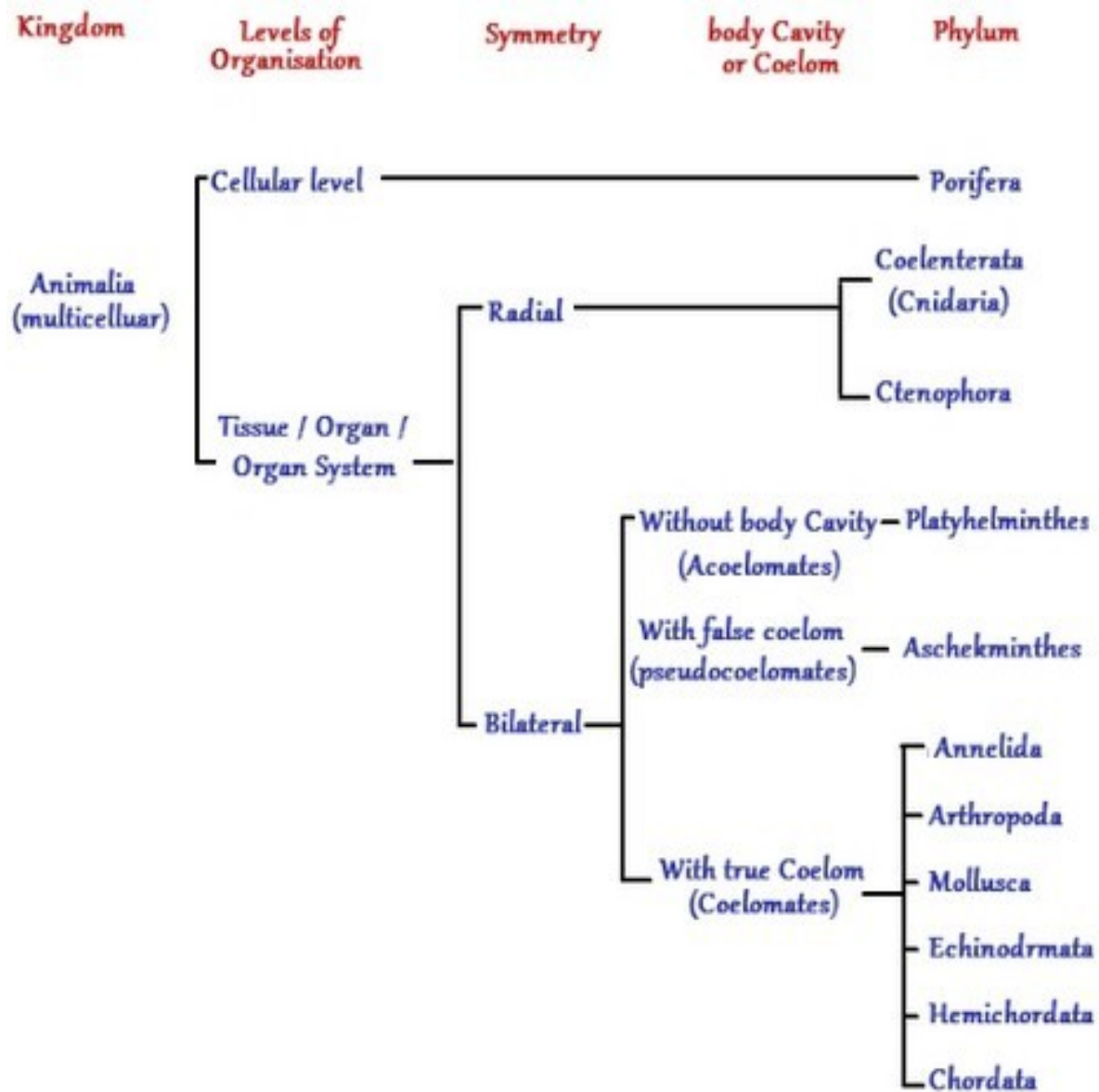
Levels of Organisation

All the organisms of the Animal kingdom are multicellular but they do not exhibit the same pattern of organization of cells.

The patterns of cellular organization seen in animals are:

- Cellular Level of Organization - In these animals the cells are arranged in the form of loose cell aggregates. This kind of cellular organization is seen in sponges. Example: *Sponges*.
- Tissue Level of Organization - Cells of the animals show division of activities among themselves. Cells performing the same function are arranged as tissues. Example: *Coelentrates*.
- Organ Level of Organization - Tissues of the animals performing the similar function are grouped to form organs. Each organ is specialized for specific function. Example: *Platyhelminthes*.
- Organ system Level of Organization - In animals where organs have associated to form functional systems where each system is concerned with a specific physiological function are said to exhibit organ system level of organisation. Example: *Annelids, Arthropods, Molluscs, Echinoderms and Chordates*.

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Patterns of Organ Systems

Organs systems in different groups of animals exhibit various patterns of complexities.

- *Digestive System* - There are two patterns of digestive system incomplete and complete digestive system.

Incomplete digestive system - This pattern of digestive has only one opening to the outside of the body, i.e., a single opening serves as both mouth and anus. Example: Platyhelminthes.

Complete digestive system - In this pattern there are two opening to the outside of the body, the mouth and the anus. Example: Arthropods, Chordates, etc.

- *Circulatory System* - Circulatory system may be of two types - open type and closed type.

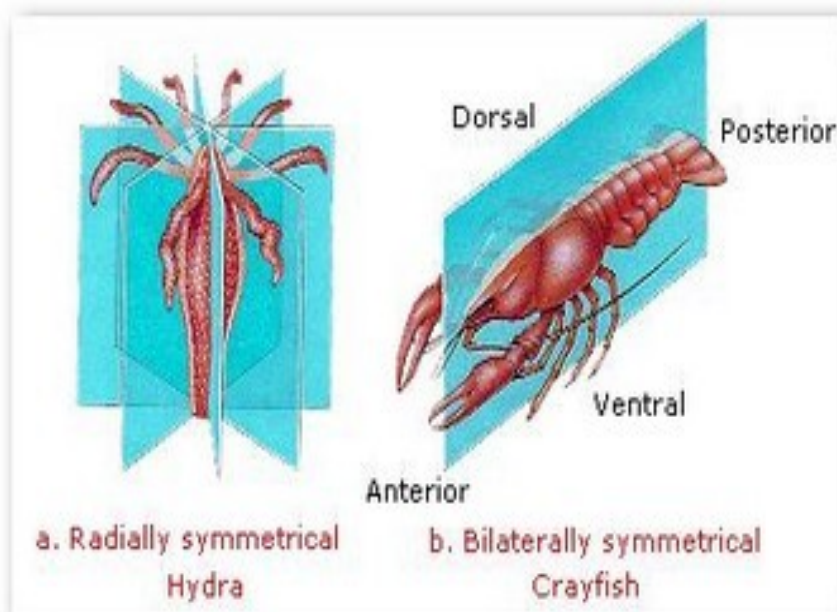
Open type - In open type circulatory system the blood is pumped out of the heart and the cells and tissues are directly bathed in it.

Closed type - In this type of circulatory system the blood is circulated through a series of vessels of varying diameters - the arteries, veins and capillaries.

Body Symmetry

Animals can be categorized on the basis of their body symmetry. The arrangement of body parts around a central point or line determines the symmetry.

- Asymmetrical - some animals cannot be divided into two equal halves along any plane passing through the center of the organism. Asymmetry is the complete absence of symmetry. Example: Sponges.
- Radial Symmetry - Animals are said to exhibit radial symmetry, when any plane passing through the central axis of the body divides the organism into two identical halves. Example: Coelentrates, ctenophores and echinoderms.
- Bilateral Symmetry - Animals where body can be divided into identical right and left halves are said to be bilaterally symmetrical. Example: Annelids, Arthropods, etc.

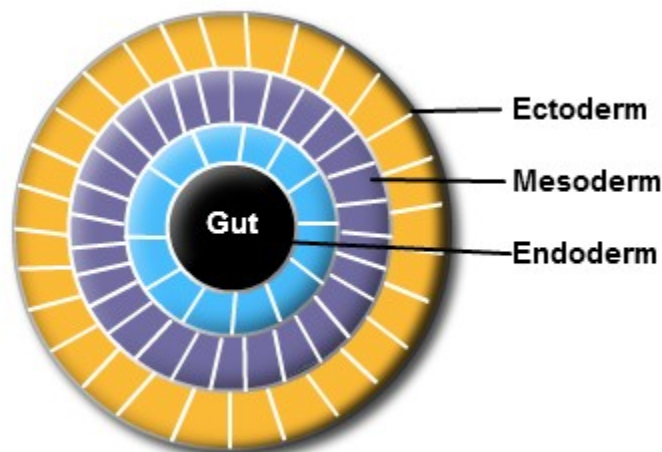
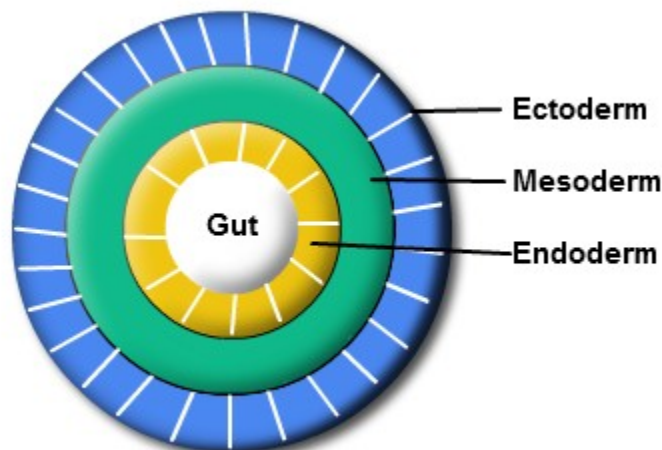


Body Wall

Body wall of animals are arranged in two or three embryonic layers. Accordingly animals are diploblastic or triploblastic.

- Diploblastic Animals - Animals in which the cells are arranged in two embryonic layers are known as diploblastic animals. Diploblastic animals have an external ectoderm and an internal endoderm. The middle mesoglea is the middle undifferentiated layer present between outer ectoderm and middle mesoderm. Example: Coelentrates.
- Triploblastic Animals - Triploblastic animals are those, whose cells are arranged in three germinal layers, the outer ectoderm, inner endoderm and the third germinal

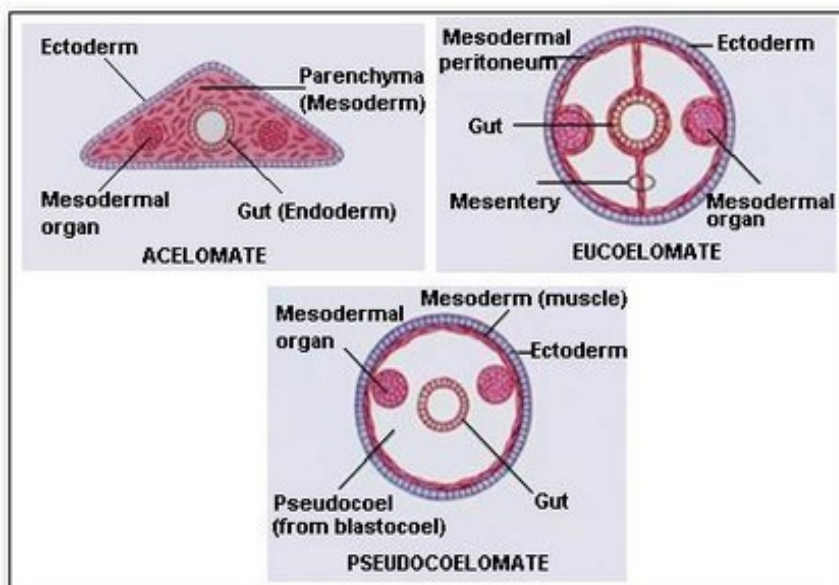
layer mesoderm which is in between outer ectoderm and the endoderm. Example: Platyhelminthes to Chordates.



Nature of Coelom

The presence or absence of a cavity between the body wall and the gut wall is important characteristic for classification. Coelom is the body cavity that is lined by the mesoderm. There are three types of animals based on the type of coelom - coelomates, pseudocoelomates or acoelomates.

- Acoelomates - Animals in which the body cavity is absent are known as acoelomates. Example: Platyhelminthes.
- Pseudocoelomates - Animals in which the body cavity is not lined by the mesoderm, instead the mesoderm is present in scattered pouches in between the ectoderm and the endoderm are known as the pseudocoelomates. Example: Aschelminthes.
- Coelomates - Animals possessing the body cavity which is lined by the mesoderm are known as coelomates. Example: Annelids, Mollusca, Arthropods, Echinoderms, chordates.



Segmentation

In some animals, body is segmented externally and internally, with serial repetition of at least some organs. This phenomenon is known as metamerism and the segmented body pattern is known as metameric segmentation. Example: Earthworm.

Notochord

In some animals during embryonic development, a mesodermally derived rod-like structure is formed on the dorsal side, this is known as notochord. Animals with notochord are known as chordates and the animals which do not form notochord are known as non-chordates. Example: Porifera to Echinoderms.

Animal Kingdom Classification System

There are many different types of animals which are similar and different to each other in many aspects. Members of a particular group of animal share a particular characteristic that is common to all the members of the group. This is the feature that defines the group.

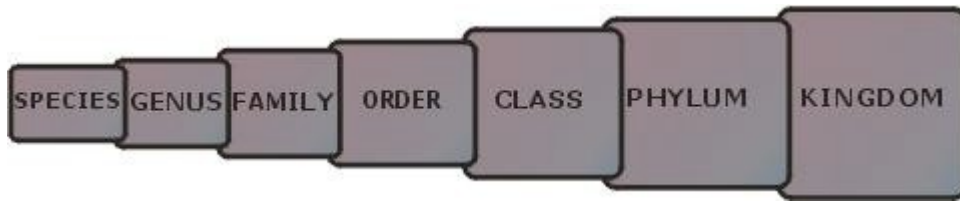
Biologists arrange organisms into groups on the basis of traits which they share with other animals and the genetic relationship with each other. This orderly form of classification of animals is the basis of *taxonomy*. Modern taxonomy is based on physical characteristics and genetic characteristics. *Systematics* is the field of study that focuses on evolutionary relationships between living organisms.

Carlous Linnaeus (1707-1778), a Swedish botanist was the inventor of modern scientific classification. He designed his system of classification so that each animal and plant he described had only one name and this name would not be shared with any other organism.

The system most scientist use put each living thing into seven groups or taxons. They are organized from most general to most specific category. These categories in the

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hierarchical system are from higher and most inclusive to lower to more specific are *kingdom, phylum, class, order, family, genus and species*.



Kingdom is the highest primary division in which all objects are placed. All animals are part of the Animal Kingdom.

Phylum - Each kingdom is divided into smaller units called phyla. Example, chordates are a phylum with members possessing the nerve cord.

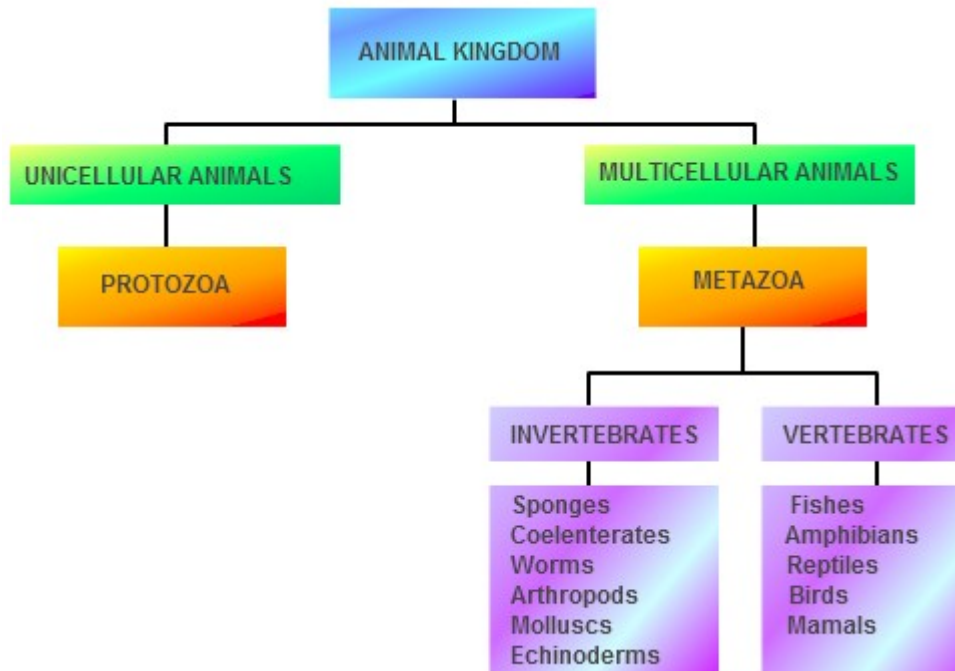
Class - The chordates are further divided into classes such as Mammalia, Birds, Reptilia, Amphibians. Members of each class have a characteristics that they share with the members of the same class but are not found in members of other classes.

Family - Classes are further divided into families. Families contain more than one genus.

Genus - Families are sub-divided into genera. Animals that share the same genus are very similar and probably evolved from the same common ancestor.

Species - Species is the most fundamental and contains single type of animal.

Animal Kingdom Classification Chart



Animals are classified into two principal groups *invertebrates* (with-out backbone) and *vertebrates* (with backbone). Backbone is the observable feature which defines whether the animal is vertebrate or

invertebrate.

Sponges, star fish, Obelia, worms, spiders, insects are the sub-groups of the invertebrate group, they do not have a backbone.

Fishes, birds, frogs, snakes and mammals have a backbone and are the sub-group of the vertebrate group.

Invertebrates

Invertebrates are the most abundant organisms on earth. They occupy almost all habitats, they can be found crawling, flying, swimming or floating. Invertebrates are the animals without backbone. These animals do not have internal skeleton made of bone. They play a vital role in the earth's ecosystem.

About 99 per cent of the known organisms are invertebrates. Out of the planets estimated 15-30 million species about 90% of the animals are invertebrates. These come in many shapes and sizes and provide services that are vital for our survival.

The most common invertebrates include sponges, annelids, echinoderms, molluscs and arthropods. Arthropods includes insects, crustaceans and arachnids.



The earliest fossil of an invertebrates dates back towards the late Precambrian period, over 600 million years back.

What are Invertebrates?

Invertebrates are the animals that do not have a backbone or the vertebral column. Animals without the notochord are invertebrates. Most of the animals are invertebrates. The term invertebrates is a prefixed form of a *Latin* derived word 'Vertebra'. 'Vertebra'

means joint in general, specifically it means '*the joint of the spinal column of the vertebrate*'. It is coupled with the prefix "in" meaning *not* or *without*, which conveys the meaning '*those that lack vertebrae*'.

Invertebrates are the most diverse group having about 12 million live species. Most of the animals on earth are invertebrates. They are cold-blooded animals; their body temperature depends on the temperature of the atmosphere.

Characteristics of Invertebrates

General characteristics of invertebrates are as follows:

- The main characteristic that separates invertebrates from other organisms is the absence of the spinal column and backbone.
- They are multicellular organisms, they completely lack cell walls.
- They are devoid of a hard bony endoskeleton.
- Due to the lack of complex skeletal systems, some invertebrates tend to be slow and small in nature.
- Due to the lack of the backbone and complex nervous system the invertebrates cannot occupy multiple environments, though they are found in the harshest of the environments.
- Invertebrates live all over the world in various habitats.
- Body is divided into three parts - head, thorax and the abdomen.
- They do not have lungs for respiration.
- Respiration is through skin.
- Some invertebrate groups possess a hard, chitinous exoskeleton.
- Most of them have tissues, that are specific organization of cells.
- Most of them reproduce sexually by the fusion of the male and female gametes.
- Few invertebrates like the sponges are sedentary, but most of the organisms are motile.
- Most invertebrates are organized with symmetric body organization.
- They can not make their own food, are heterotrophs.

List of Invertebrates

Invertebrates are the most diverse organisms present on earth. Almost 95% of the animal populations are of the invertebrates. Based on the International Union regarding Conservation of Nature at the time of 2009 more than 1.3 million invertebrates were identified. Invertebrates make up around 75% of the recognized species on Planet.

The actual number of invertebrates is unknown, there are several predictions that there may be tens of millions of invertebrates, majority being the insects. New species of invertebrates are being discovered regularly, another worrying fact is that there is insufficient information about these organisms, the invertebrates might be going extinct and the scientists would never know that they even existed.

Below are a list of invertebrates -

Crustaceans, Centipedes, Ants, Wasps, Spiders, Locusts, Honey bees, Termites, Cockroach, Grasshoppers, Crickets, Stick insect, Mantis, Crabs, Sponges, Star fish, Unio, Leeches, Earthworms, etc.

Invertebrates Classification

Some invertebrate phyla are:

Phylum Porifera (Sponges)

- Members of this phylum are commonly known as *sponges*.
- Habitat - They are mostly marine, few are found in fresh water.
- Body symmetry - mostly are *assymetrical animals*, no definite shape to the body.
- Level of organization - These are *primitive* animals, multicellular with *cellular grade* of organization.
- Motility - Adult sponges are *sessile*, that is they need a substratum to attach themselves to a surface and do not move.
- Mode of Nutrition - Due to the sessile nature, sponges are *filter feeders*.
- Digestion - Digestion is *intracellular*.
- Skeleton - The body of sponges is supported by a skeleton made of *spicules* or *spongin fibres*.
- Reproduction - Sexes are not separate, they are *hermaphrodites*. Hermaphroditism - condition where eggs and sperms are produced by the same individual. Sponges reproduce asexually by fragmentation and sexually by formation of gametes.
- Fertilization - Fertilization is *internal*.
- Development - *Indirect* development, having a larval stage which is morphologically distinct from the adult.
- Water transport or Canal system - Sponges have water transport system. Water enters through minute pores (ostia) in the body wall into a central cavity known as spongocoel. From the spongocoel water goes out through the osculum. This water system aids in food gathering, respiratory exchange and removal of wastes.
- Choanocytes - These are collar cells, they line the spongocoel and the canals.
- Examples: *Sycon*, *Euspongia*, *Spongilla*.



Phylum Cnidaria (Coelentrata)

- Habitat - They are *aquatic* mostly marine animals, they are *sessile* or *free-swimming*.
- Level of organization - They exhibit *tissue level* of organization.
- Body wall - They are *diploblastic* animals, body wall is made of 2 layers, outer ectoderm and inner endoderm.

- Body symmetry - They are *radially* symmetrical.
- Digestive system - They have a central gastro-vascular cavity with a single opening *hypostome*, which serves as both the mouth and the anus.
- *Digestion* is extracellular and intracellular.
- Specialized cells known as the *cnidoblasts* or *cnidocytes* which contain the stinging capsules or nematocytes are present on the tentacles and the body. Cnidoblasts are used for anchorage, defense and for the capture of prey.
- Some cindarians like corals have skeleton composed of calcium carbonate.
- Cnidarians exhibit two basic body forms called *polyp* and *medusa*. The polyp is a sessile form ann cylindrical in shape like the Hydra, Adamsia etc. The medusae form is umbrella shaped and free-swimming forms like Aurelia or jelly-fish.
- Metagenesis - Cnidarians which exhibit both polyp and medusae form are exhibit alteration of generation, this is known as metagenesis. Poplyps produce medusae asexually and medusae form the polyps sexually.
- Locomotion - The body contains *nerve network* that allows movement of tentacles and body.
- Examples: Aurelia (medusa), Adamsia (polyp).



Phylum Ctenophora :

- Ctenophores are commonly known as *sea walnuts* or *comb jellies*.
- Habitat - They are exclusively *marine* animals.
- Body symmetry - They are radially symmetrical.
- Body wall - They are *diploblastic* organisms.
- Level of organization - Ctenophores exhibit *tissue level* of organization.
- Digestion is both extracellular and intercellular.
- Locomotion - The body bears eight external rows of *ciliated comb plates*, these help in locomotion.
- *Bioluminescence* is a property of living organism to emit light is well-marked in ctenophores.
- Reproduction - Sexes are not separate, reproduction is by sexual means only.
- Fertilization is *external* with indirect development.
- Example: *Pleurobrachia*

Phylum Platyhelminthes

- They are commonly called as flatworms, they have *dorso-ventrally* flattened body.
- Habit - They are mostly *endoparasites* found in animals and human beings.
- Habitat - Fresh-water and salt water; terrestrial.
- Body symmetry - Flatworms are *bilaterally* symmetrical.
- Body wall - They are *triploblastic* animals.
- Coelomic cavity - They are *acoelomate* animals
- Level of organization - *Organ level* of organization.
- Digestive system - In flatworms digestive system is *incomplete*, that is the digestive cavity has a single opening.
- Parasitic flatworms possess hooks and suckers to hold on to the body of the host. Some forms absorb nutrition directly from the host, through their body surface.
- Osmoregulation and excretion is carried out by specialized cells called *flame cells*.
- Sexes are not separate.
- Fertilization is internal and development is indirect, having many larval stages.
- Some members like Planaria possess high *regeneration capacity*.
- Examples: *Taenia* (tapeworm), *Fasciola* (liver fluke).



Phylum Aschelminthes (Nematoda)

- The body of worms of aschelminthes in cross-section is circular, hence the name round worms.
- Habitat - They may be free-living, aquatic and terrestrial or parasitic in plants and animals.
- Level of organization - Round worms have *organ-system level* of organization.
- Body symmetry - They are *bilaterally* symmetrical.
- Body wall - They are *triploblastic* animals.
- Coelomic cavity - They are *pseudocoelomate* animals.
- Digestive system - This is the first phylum to have a *complete digestive system*, with a well developed muscular pharynx.
- Excretory system - An excretory tube removes body wastes from the body cavity through the excretory pore.
- They are *dioecious* - the sexes are separate i.e., males and females are distinct. Often females are longer than the males.
- Fertilization is internal.
- Development may be direct - the young ones resemble the adult, or indirect.
- Examples: *Ascaris* (round worm), *Wuchereria* (filaria worm), *Ancylostoma* (hookworm).



Phylum Annelida (Segmented worms)

- Habitat - They may be *aquatic* either marine or fresh water; or terrestrial; free-living and sometimes parasitic.
- Level of organization - They exhibit *organ-system level* of body organization and bilateral symmetry.
- Body wall - They are *triploblastic*.
- Coelom - They are coelomate animals.
- Body is *metamerically* segmented. The body surface is distinctly marked out into segments of metameres and hence, the phylum name Annelida.
- Locomotion - They possess longitudinal and circular muscles which help in locomotion.
- Aquatic annelids possess lateral appendages, *parapodia* which help in swimming.
- Circulatory system is closed.
- Osmoregulation and excretion is by *Nephridia*.
- Neural system - It consists of paired ganglia connected by lateral nerves to a double ventral nerve cord.
- *Nereis* is *dioecious*, but earthworm and leeches are monoecious.
- Reproduction is sexual.
- Examples: *Nereis*, *Pheretima* (earthworm), and *Hirudinaria* (blood sucking leech)



Phylum Arthropoda

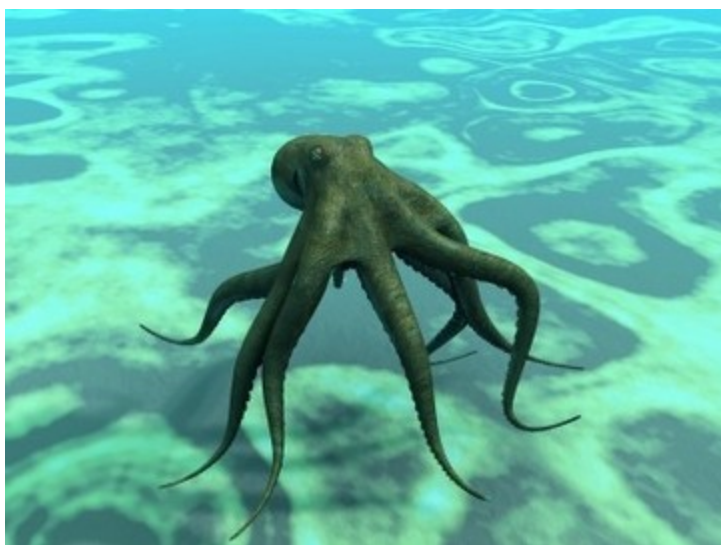
- It is the largest phylum of the Animalia.
- It includes insects, spiders, crayfish, etc.
- Level of organization - They have *organ-system level* of organization.
- Body symmetry - They are *bilaterally* symmetrical.
- Body wall - *triploblastic*, segmented. Coelomate animals.
- The body of arthropods is covered by *chitinous exoskeleton*.
- Body is divided into head, thorax and abdomen.
- *Jointed appendages* - arthros - jointes, poda - appendages, hence the name is derived from this characteristic.
- All the arthropods have jointed appendages which give arthropods a wide range of controlled motions.
- Respiration is through organs like gills, book gills, book lungs or tracheal system.
- Circulatory system - It is of open type. Sensory organs are present, antennae, eyes (compound and simple), statocysts or balance organs are present.
- Excretion - It takes place through *malphigian tubules*.
- Mostly they are *dioceious*. Fertilization is usually internal.
- They are *oviparous* animals.
- Development may be direct or indirect.
- Examples: Honey bee, Silkworm, Lac insect, Mosquitoes, Locust, Crab



Phylum Mollusca

- It is the second largest phylum.
- Habitat - Molluscs are terrestrial or aquatic; they may be marine or fresh water.
- Level of organization - They have an *organ-system level* of organization.

- Body symmetry - *Bilaterally* symmetrical.
- Body wall - *triploblastic*. Coelomate animals.
- Body is covered by a *calcareous shell*.
- Body is unsegmented, they have a distinct head, muscular foot and visceral hump.
- The *radula* - Mouth of the molluscs contain tongue-like organ called radula, which has many rows of teeth, which is used to scrape food.
- *Mantle* - It is a fold of skin that surrounds the body organs.
- It is a soft and spongy layer of skin that forms a mantle over the visceral hump.
- The space between the hump and the mantle is called the *mantle cavity*.
- Feather like gills are present in this cavity.
- These gills have respiratory and excretory in functions. Anterior head region has sensory tentacles.
- Example: Pila, Octopus, Pearl oyster, Loligo, Sea-hare, Chiton.



Phylum Echinodermata

- All the members are marine, live mainly on the ocean floor.
- These animals have an endoskeleton of *calcareous ossicles*, and hence the name echinodermata (spiny bodied).
- Level of organization - *Organ-system level* of organization.
- Body symmetry - The adults are *radially* symmetrical, but the larvae are bilaterally symmetrical.
- Body wall - *Triploblastic*. Coelomate animals.
- Digestive system is complete.
- The mouth is present on the ventral side and anus on the dorsal side.
- The most distinctive feature is the *water vascular system*.
- This helps in locomotion, capture and transport of food and respiration.
- Excretory system is absent.
- Sexes are separate.
- Reproduction is by sexual means.
- Fertilization is usually external.
- Development is indirect with free-swimming larva.
- Example: Star fish, Sea urchin, Brittle star.



Examples of Invertebrates

Invertebrates include all animals that do not belong to the phylum chordata. Common examples include clams, snails, spiders, cockroach, worms, star fish, octopus.

Porifera - Sponges, *Sycon* (scypha), *Spongilla* (fresh water sponge) and *Euspongia* (bath sponge).

Cnidaria - *Aurelia*, *Adamis*, *Hydra*, *Sea anemones*, *Physalia* (Portuguese man-of-war), *Pennatula* (sea-pen), *Gorgonia* (sea-fan), *Meandrina* (brain coral).

Ctenophora - *Pleurobranchia* and *Ctenophora*.

Platyhelminthes - *Taenia* (tapeworm), *Fasciola* (liver fluke).

Aschelminthes - *Ascaris* (round worm), *Wuchereria* (filaria worm), *Ancylostoma* (hookworm).

Annelida - *Nereis*, *Pheretima* (earthworm) and *Hirudinaria* (blood sucking leech).

Arthropoda - *Apis* (honey bee), *Bombyx* (silkworm), *Laccifer* (lac insect); Mosquitoes - *Anopheles*, *Culex* and *Aedes*; *Locusta* (locust) ; *Limulus* (king crab).

Mollusca - *Chaetopleura* (Chiton), *Loligo* (squid), *Pila* (apple snail), *Pinctada* (pearl oyster), *Sepia* (cuttlefish), *Loligo* (squid), *Octopus* (devil fish), *Aplysia* (sea hare), *Dentalium* (tusk shell).

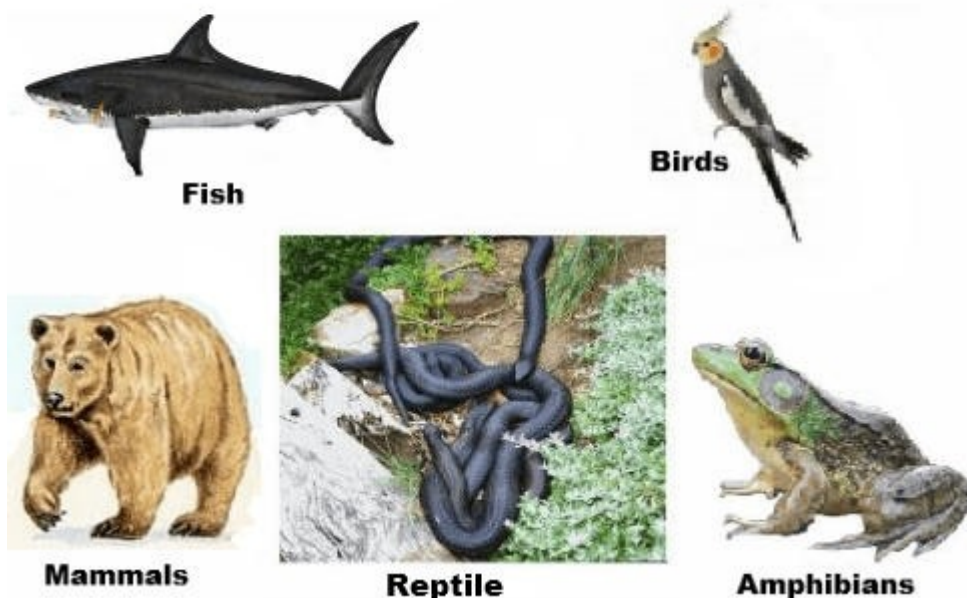
Echinodermata - *Asterias* (star fish), *Echinus* (sea urchin), *Antedon* (Sea lily), *Cucumaria* (sea cucumber), *Ophiura* (brittle star).

Vertebrates

Vertebrates are the most organized organisms on Earth. They belong to the sub-phylum Vertebrata. They are not the most numerous group of animals, they are the most advanced group of animals. The characteristics that makes vertebrates special are the presence of the spinal cords, vertebrae and notochords.

Vertebrates

Animals with backbones



Most vertebrates have a very well developed nervous system. The vertebrates also have muscles and skeletons which help them move around efficiently and perform complex moves.

Vertebrates include the majority of the Phylum Chordata, having about 64,000 species described. Vertebrates make about 4% of all described species.

Vertebrates have a long history on earth - more than 500 million years, from the Cambrian era until date. First Vertebrates are said to have appeared around the Cambrian period of the Paleozoic era about 525 million years ago, they are said to be adapted to feeding on algae in shallow waters and moving around from place to place.

Vertebrates Definition

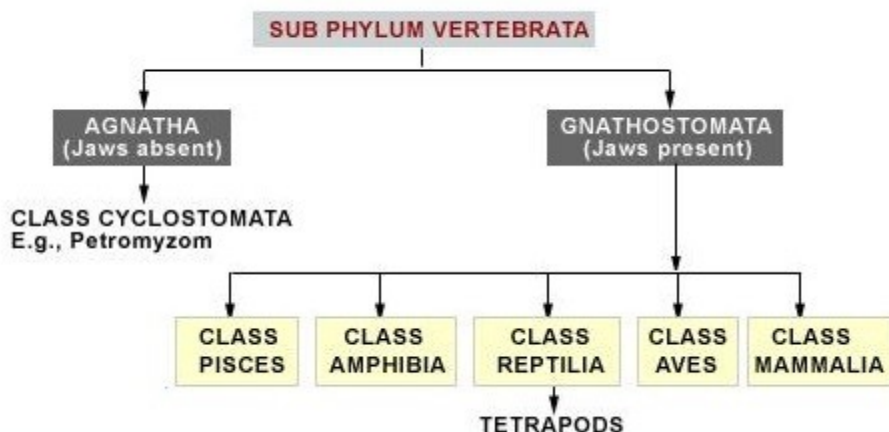
Vertebrates are the animals belonging to the sub-phylum Vertebrata. They belong to the Phylum Chordata. The characteristics of phylum chordata is the presence of notochord, a dorsal hollow nerve cord and paired pharyngeal gill slits. The members of subphylum Vertebrata possess notochord during the embryonic period. The notochord is replaced by a cartilaginous or bony vertebral column in the adult. All vertebrates are chordates but all chordates are not vertebrates. Special characteristics of vertebrates other than the vertebral column are, they have a muscular heart which is two, three or four chambered. For excretion they have kidneys and appendages that are paired which may be fins or limbs.

Vertebrate Characteristics

General characteristics of the sub-phylum Vertebrata are as follows:

- Vertebrates have a well developed brain.
- Brain is set inside a bony box, known as cranium.
- Notochord is present in all chordates during sometime of development, it is formed on the dorsal side of the primitive gut in the early embryo stage.
- Most vertebrates possess guts with a non-terminal anus.
- The mouth cavity and the oesophagus is connected by the pharyngeal gill slits to a muscular tube pharynx, which opens to the exterior.
- They possess a dorsal hollow nerve tube at some stage of their life.
- Possess a dorsal cartilaginous nerve rod known as the notochord.
- At some stage of their life possess gill slits in the pharyngeal region.
- Have partially open circulatory system.
- Possess two pairs of appendages.
- The endoskeleton is made of cartilage or bone.
- The first vertebrates were jawless fishes with single caudal fin.
- The advancement of vertebrates with a hinged jaw which opened new food options and jawed fishes became the dominant creatures in the sea.
- All vertebrates have a heart and closed circulatory system.
- Reproduction is normally sexual.
- Feed on variety of organic materials.
- Unisexual animals, have one pair of gonads.

Vertebrate Classification



Classes of Vertebrates

Class - Cyclostomata

The living members of this class are all ectoparasites on some fishes. They have a elongated body. They bear 6-15 pairs of gills through which they respire. The mouth of the cyclostomes is sucking circular mouth without jaws. They do not have body scales and paired fins. The vertebral column and the cranium is cartilaginous. Circulation is closed type. These are marines organisms but they migrate to fresh water for spawning. After

metamorphosis their larvae returns to the ocean. Example: Petromyzon (Lamprey), and Myxine (Hagfish).

Class - Chondrichthyes

These organisms are marine and have streamlined body. The endoskeleton is cartilaginous. Mouth is located ventrally and the notochord is present throughout life. The gills are separate and are not covered by the operculum. The skin contains minute placoid scales. The placoid scales are modified as teeth and the jaws are powerful. They are predaceous animals. Air bladder is absent in these animals, hence, they have to swim constantly to avoid shrinking. Example: Scolidon, Pristis, Trygon.



Class - Osteichthyes

This class includes animals that are both marine and fresh water fishes with bony endoskeleton. Their body is streamlined. Mouth is terminal in position. The gills are covered with operculum on each side and are in four pairs. The skin is covered by ctenoid/ cycloid scales. Air bladder is present, it regulates buoyancy. Two-chambered heart is present, with one auricle and one ventricle. They are poikilothermic animals. The sexes are separate, fertilization takes place externally. Most of them are oviparous animals and development is direct, with no larval stages. Example: Hippocampus, Clarias.



Class - Amphibia

The name Amphibia indicates - from Greek, *Amphi* meaning dual and *bios* meaning life. Amphibians can live both in aquatic and terrestrial habitats. These animals have two pairs of limbs. Body is divisible into head and trunk and tail is present in some animals. The skin of these animals are scaleless and moist. The eyes have eyelids, and the ears are represented by a tympanum. Cloaca is a opening to the exterior, it is a common chamber for the alimentary canal, urinary and reproductive tracts. Gills, lungs and skin aids in respiration. Heart is three-chambered. They are cold-blooded animals. Sexes are separate, fertilization is external. They are oviparous animals and development is indirect.

Example: Toad, frog, salamander.



Class - Reptilia

In Latin *reperere* or *reptum* means to creep or crawl, hence, the class name refers to locomotion that is of creeping or crawling mode.

These animals are mostly terrestrial and their body is covered dry and cornified skin, epidermal scales or scutes. External ear opening is absent, tympanum represents the ear. Limbs, if present are of two pairs. Usually heart is three-chambered, but is four chambered in crocodiles. They are poikilothermic animals. Some animals like the snake and the lizards shed their skin. Sexes are separate, internal fertilization takes place. They are oviparous and development is direct.

Example: Turtle, Chameleon, crocodile.



Class - Aves

Most of the members can fly, except the flightless birds. The characteristic feature of birds is the presence of *feathers*. The forelimbs are modified as wings. The hind limbs are modified for walking, swimming or clasping and generally have scales. The skin is dry and does not have glands, except a oil gland at the base of the tail. Endoskeleton is bony, and the bones are hollow with air cavities known as *pneumatic bones*. Heart is four-chambered completely. They are *warm-blooded* animals. Lungs are the organs of respiration. Sexes are separate, fertilization is internal. They are *oviparous* animals and development is direct.

Example: Crow, Pigeon, Parrot etc.



Class - Mammalia

Mammals are present in almost all habitats - polar ice caps, deserts, mountains, forests, and grasslands. The unique characteristic of the class mammalia is the presence of milk producing glands (*mammary glands*), by which the young ones are nourished. The limbs are of two pairs. The skin is covered with hairs. External ear '*pinna*' is present. Heart is four-chambered and they are homeothermic animals. Respiration is through lungs. Sexes are separate and fertilization is internal. They *viviparous* animals, with direct development.

Example: Kangaroo, Tiger, Lion, Platypus etc.

List of Vertebrates

The vertebrates are a very diverse group of animals ranging from Hag-fish to Man. There are not the most numerous group of animals. Vertebrates are interesting because we are intimately familiar with them, or species *Homo sapiens* is included within the sub-phylum vertebrata. Vertebrates are easy to find as we co-exist with them and often aware of their presence in our environment.

Below are the list of few very familiar vertebrates: deer, fish, dogs, cats, birds, snakes, lizards, kangaroo, bears, foxes, wolves, bat, monkeys, eagles, penguins etc.

Examples of Vertebrates

Below are the examples of vertebrates:

Pisces: Sharks, trouts, eels, tunas, seahorse, pirahnas, salmons, etc.

Amphibians: Salamanders, frogs, toad, newts, caecilians, etc.

Reptiles: Snakes, lizards, tortoise, turtles, crocodiles, gharials, alligators, etc.

Birds: Penguins, emu, eagles, ostrich, parrot, crow, pigeon, ducks, owls, kites, robins,

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woodpeckers, albatross etc.

Mammals: Kangaroo, echidna, platypus, bats, mice, tigers, moles, dolphins, whales, deer, gorillas, lemurs, rabbits, wolves, lions, leopards, jackals, etc.

SOCIO-ECONOMIC IMPORTANCE OF KINGDOM ANIMALIAE

- **Tourism:** Tourists visits to game parks boost the economy and create jobs.
- **Food:** e.g. Cows and goats are raised for meat and milk.
- **Hunting** provides food and income from selling game meat.
- **Leather:** Hides of mammals are used to make leather products e.g. shoes, belts and bags.
- **Fishing** provides food and jobs.
- **Seed dispersal** e.g. fruit bats.
- **Pollinating agents:** Animals such as insects (bees) and birds are pollinating agents in plants and crops.